

Optical Integrated Sensing and Communication for 6G: A Systematic Review and Survey of Architectures, Metrics, and Tradeoffs Across Optical Platforms

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Abstract—Light carries data and signatures of the path and objects it encounters, allowing the same optical infrastructure to support communication and sensing. Optical integrated sensing and communication (O-ISAC) builds on this dual capability across fiber, free space optical, visible light and LiFi, millimeter wave and terahertz systems enabled by photonics, and hybrid systems. This breadth raises a central question: which design insights are shared, and which depend on a particular platform? A systematic literature search through 22 June 2026 identified 227 eligible reports published from 2020, corresponding to 206 unique studies. Using structured narrative synthesis and a review specific technical assessment, we organized the evidence by physical modality and by how communication and sensing are coupled. The review received no specific support and was retrospectively registered on OSF (7f6wb; DOI: 10.17605/OSF.IO/7F6WB). Our comparison framework keeps each result connected to its task, measurement plane, operating conditions, and validation setting. The evidence reveals conditional relations in which bandwidth, optical power, hardware constraints, and signal design jointly shape communication reliability and sensing performance, extending design reasoning beyond a single rate and resolution frontier. Laboratory demonstrations already span a wide range of architectures; field validation, shared benchmarks, and reusable data and code offer the next stage of maturity. Together, the taxonomy and framework help researchers interpret evidence, choose meaningful benchmarks, and advance O-ISAC toward 6G.

Index Terms—6G, integrated sensing and communication, metric comparability, optical communication, optical sensing.

I. INTRODUCTION

An optical link is designed to carry information, yet the received optical field also carries signatures of its interaction with the surrounding world. Motion can alter phase or induce

a Doppler shift, while reflected paths encode delay and received power; turbulence and strain leave different signatures. When these changes are explicitly interpreted as environmental observables rather than treated only as channel effects, the optical system can also serve as a sensor. This principle already appears in distributed sensing over fiber, positioning based on visible light, and turbulence-aware optical links [1]–[3].

The appeal of this approach becomes clear at the system level: communication networks already provide optical sources, transmission paths, receivers, and signal processing that may also support sensing. Reuse, however, couples the two functions. Resources used to preserve sensing observability can constrain payload transmission, while a receiver designed only for communication may not retain the phase, delay, or spatial information that sensing requires. The engineering question is therefore not simply whether an optical system can support both functions, but what can be shared and which constraints follow from that sharing.

This question has gained importance as integrated sensing and communication (ISAC) has been identified as an IMT-2030 usage scenario [4]. Optical and photonic systems extend this agenda into physical regimes that differ from conventional radio frequency (RF) formulations. Guided fiber provides spatially constrained interaction paths and several optical observables, whereas free space optical (FSO) systems can combine a directional communication path with lidar but remain sensitive to alignment and atmospheric propagation [5]. Systems based on visible light face their own constraints, including blockage, background light, and illumination requirements. Photonics can generate, distribute, and process wideband signals in the millimeter wave and terahertz bands [6], [7], although the radiated path remains RF. Hybrid designs can connect more than one of these regimes. Optical ISAC (O-ISAC) is therefore best understood as a family of related architectures rather than a single interchangeable platform.

That diversity requires an explicit survey boundary. Here, O-ISAC covers optical or photonic systems in which communication and sensing are considered together, integrated, jointly designed, jointly optimized, or jointly evaluated within a common technical setting. That setting may lie in the system architecture, optical link, waveform or resources, hardware, channel model, or application scenario. Sharing a waveform and operating concurrently are particular forms of coupling rather than prerequisites for inclusion.

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Placing these systems within the same scope, however, does not make their reported results directly comparable. Familiar labels can conceal important differences: theoretical resolution should not be treated as measured estimation error, nor should the throughput of a single channel be treated as aggregate capacity. We treat numerical comparisons across studies as defensible only when the metric definition, measurement plane, operating conditions, any relevant baseline, and validation setting are explicit.

A. Prior Syntheses and Survey Positioning

Existing surveys explain complementary parts of this field. Some center on O-ISAC architectures or analytical limits [8], [9]; others place optical systems within the wider evolution of ISAC [10] or examine particular media and developments across several platforms [11]–[14]. These perspectives differ not only in topic but also in the object they synthesize. Reading them by the question they help a reader answer therefore clarifies what is already available and what comparison task remains. Table I groups the surveyed literature by reader task, evidence unit, and comparison logic.

Taken together, the six families provide depth in architectures, models, physical media, photonic implementations, and deployment agendas. Their evidence units differ, however: a relation defined within a model, an observation native to a medium, and an architecture or deployment object support different kinds of reasoning. None is intrinsically stronger, but the O-ISAC label alone supplies no common denominator. The remaining comparison task is therefore to preserve each result's native context and state explicitly when a relation across studies is conditional or merely descriptive.

This article is a survey of O-ISAC evidence at this boundary. We mapped 227 eligible reports to 206 unique studies, linking companion reports to one study record rather than treating them as independent demonstrations. The survey therefore complements reviews of architecture, theory, and individual media rather than placing platforms on a universal scale. The primary evidence window extends from 1 January 2020 to 22 June 2026.

The review asks how optical platforms and coupling mechanisms are organized, how consistently communication and sensing metrics are reported and when they can be compared, which conditional tradeoffs recur, which technologies and applications shape current systems, and what validation, benchmark, and reproducibility gaps constrain progress toward 6G. These linked questions combine the seven protocol questions into the four reader tasks addressed by the technical synthesis.

On this basis, this survey makes four contributions:

- We develop a taxonomy that spans optical platforms and distinguishes physical modality from the mechanisms used to couple communication and sensing.
- We introduce a comparison framework that preserves measurement context and separates interpretation within individual studies, conditional comparison across studies, and descriptive use when direct comparison is not defensible.

- We synthesize 402 substantive tradeoff records from 168 studies, group them by recurring engineering mechanisms, and show why the evidence does not support a universal relation between communication rate and sensing resolution across optical platforms.
- We assess validation maturity, benchmark readiness, and reported data and code availability, then use the observed gaps to identify priorities for comparable experiments, reusable research artifacts, and validation under deployment conditions.

The remainder of the survey follows the same analytical sequence. Section II establishes the system boundary, technical vocabulary, and comparison framework. Section III then explains how the evidence was identified, governed, and appraised, and reports the resulting evidence base. Section IV examines optical platforms together with their integration architectures, whereas Section V relates performance metrics to recurring joint design tradeoffs. Section VI assesses validation, reproducibility, and benchmark readiness. Section VII connects the technical findings to enabling technologies, applications, and 6G implications. Section VIII draws out the research roadmap and review limitations, and Section IX concludes the survey.

II. O-ISAC FOUNDATIONS AND COMPARISON FRAMEWORK

Optical integrated sensing and communication (O-ISAC) names a design relationship rather than a single transceiver class. Communication and sensing may meet in the propagation path, at the source or receiver, through a waveform or resource, or within a shared model or application. The integration question is common to these arrangements, but their physical models and natural performance scales need not be.

This section establishes the vocabulary needed to preserve those differences. It begins with the system boundary and the modality of each signal path, then locates the mechanisms that couple communication and sensing. Measurement plane and provenance complete the context, allowing the survey to represent a result as a comparison record rather than as an isolated number.

A. Technical System Boundary

Although the executed eligibility rule is reported in Section III, the technical synthesis focuses on where O-ISAC joins functions within a system. A fiber disturbance and a free space target produce different observables, while indoor devices and radio frequency targets introduce further physical regimes even when the architecture contains optical hardware. Shared waveform operation is therefore one form of O-ISAC rather than its universal definition, just as a photonic front end does not make a radio frequency propagation path optical. The survey label marks the technical relationship between the functions; the modality and signal path state what that relationship physically means.

TABLE I
PRIOR O-ISAC SYNTHESSES BY READER TASK, EVIDENCE UNIT, AND COMPARISON BOUNDARY. FAMILIES SUPPORT NAVIGATION RATHER THAN QUALITY RANKING.

Synthesis family	Sources	Optical scope	Evidence unit and comparison logic	Reader task	Boundary
Architecture and system design	[8], [15]–[17]	General O-ISAC, FSO, retroreflective links, and fiber networks	Architecture, optimization, target, or link model; taxonomy and relations within each scenario	How are communication and sensing coupled?	Relations remain architecture and assumption specific.
Analytical limits and estimation methods	[9], [18]	General optical systems and polarization sensing in coherent fiber links	Model, bound, objective, estimator, or learning method; within-model and method comparison	What limits follow from the assumptions?	One model does not define a modality-wide empirical scale.
Technological evolution, standardization, and deployment	[10], [19]–[22]	Radio-to-optical evolution; terrestrial and submarine networks	Historical phase, standardization path, network architecture, or deployment; synthesis by period and ecosystem	How has O-ISAC moved toward deployment?	A shared trajectory does not make infrastructures commensurate.
Physical media and optical networks	[11]–[13], [23]–[25]	Visible light LiSAC, guided fiber systems, and optical networks	Medium, observable, sensing path, or network configuration; taxonomy within related solution classes	How does each path support sensing?	Transfer requires aligned media, observables, paths, and conditions.
Waveforms, resource multiplexing, and photonic hardware	[26]–[31]	Fiber, microwave photonics, and photonics-enabled millimeter wave or THz	Waveform, multiplexing, front end, or processing chain; mechanism taxonomy with selected demonstrations	Which mechanisms enable joint operation?	Shared photonics does not equate paths, observables, or metrics.
Applications and challenges across platforms	[14]	VLC, FSO, fiber, photonics, and hybrid settings	Application, use case, solution class, or challenge; cross-platform thematic synthesis	Which applications and gaps recur?	Breadth connects research lines without making results commensurate.

Note: The 24 displayed syntheses are assigned once; electronic Table ST-RS1 retains overlapping scopes, secondary tags, and 14 short or focused syntheses (38 total). All are contextual and outside the 206-study primary base. Seven manuscript-stage additions do not alter the review flow. Mohsan et al. was captured by the executed search, but its September 2026 issue date is not used to establish precedence because first-online publication was unverified.

B. Modality and Signal Path Boundaries

We organize the evidence into six analytical families: fiber and free space optical (FSO) systems form two families, while visible light communication and LiFi (VLC/LiFi) form a third. Photonic THz, hybrid optical, and other optical systems complete the taxonomy. These review categories are not classes defined by standards; they place each result in the correct physical regime before any performance comparison is attempted.

Fiber systems place sensing in a guided path, whereas FSO systems place target geometry and beam control in a directional optical path. VLC/LiFi adds illumination and access constraints. In photonic THz systems, optical hardware may generate, distribute, or process the carrier, but the target facing path remains radio frequency. Hybrid systems require both domains and their interface to be retained. These distinctions establish the physical reference frame; Section IV later synthesizes their study counts, implementations, and platform specific constraints [2], [7], [32]–[34].

C. Integration Mechanisms and Measurement Planes

We distinguish several coupling locations without treating them as a sequence. At the broadest level, communication and sensing contribute to one application while their resources may remain separate. A shared link or channel gives both

functions a common physical path or state. Other designs share hardware or an optical carrier and may also share a waveform or resource. The resource may be time or frequency. Other systems allocate power, polarization, code, or space. Joint design or optimization is present when at least one variable is chosen in view of both communication and sensing outcomes.

These locations are neither an eligibility rule nor a maturity scale. A study may occupy several locations or combine shared hardware with separate waveforms; an abstract joint optimization need not demonstrate a shared prototype. A shared hardware application [35], a fiber system with a shared link [36], and a design that combines multiple mechanisms [32] consequently answer different engineering questions. Their attributes describe coupling and must not be collapsed into a single integration score.

This distinction prevents collocation from being mistaken for simultaneous waveform reuse. It also prevents a common carrier or receiver from being mistaken for joint optimization. Later sections therefore treat integration mechanisms as overlapping attributes and examine concurrency, resource use, and validation separately.

Coupling describes where the functions meet, but it does not establish where a reported value is defined. A measurement plane supplies that second boundary by identifying where a quantity is observed in the signal chain. Design variables include carrier and waveform settings, launch power, and path

geometry. The optical link plane contains received optical power and loss as well as optical signal to noise ratio. After conversion, electrical signal quality occupies a new plane. Digital processing then produces error rate and throughput outcomes. Sensing introduces task specific quantities for geometry and motion, environmental disturbance, classification, or detection.

Similar labels or curve shapes do not make these quantities exchangeable. Optical and electrical SNR refer to different reference planes; launch and received power are not the same operating quantity. Gross line rate is not net information rate, and pre-FEC error rate is not post-FEC error rate. Range resolution, range error, precision, and a Cramér–Rao bound also answer different questions. Comparison therefore depends on the task, target, operating point, channel, and processing chain as well as the metric name.

Provenance sets a further boundary. An analytical bound, simulation, laboratory measurement, offline dataset result, and field observation carry different validity conditions. They may support a common qualitative pattern, but not a shared numerical leaderboard unless the decisive conditions align. A within study sweep can therefore reveal a useful design relationship even when its values cannot be transferred across studies.

D. The Cross Platform Comparison Record

We therefore represent a candidate comparison as a structured record rather than an isolated number. Modality and signal path establish the physical context, while the tasks and integration mechanism describe the functional relationship. The record preserves the metric’s definition, plane, and unit. It also retains the operating point and channel together with the target and constraints. Validation type, baseline, and source locator complete its provenance; a missing field is not inferred from a neighboring study.

A reported value becomes comparison evidence only as part of a complete technical record. Table II makes that record explicit and shows which missing or misaligned fields change the permitted use.

The contract separates source-bounded interpretation, conditional cross-study synthesis, and descriptive retention. Record completeness is necessary but not sufficient: a conditional relation still requires the decisive fields to align and every remaining difference to be stated. The framework therefore does not turn heterogeneous demonstrations into a universal O-ISAC Pareto frontier.

One fiber study, for example, supplied communication and localization records that supported comparison under their recorded conditions, while its other records remained conditional [37]. This decision does not make the values transferable to FSO, VLC/LiFi, or photonic THz systems; it states only what the available provenance permits. Throughout the survey, representative studies anchor patterns, tables retain the inventory, and prose states the boundary of each inference.

The framework thus assigns a distinct role to each axis. Modality and signal path provide physical context, the integration mechanism locates the coupling, and measurement

plane identifies what was observed. The comparison record then controls how evidence can be related. The next section explains how studies, reports, and claims were selected and governed under these rules.

III. REVIEW METHOD AND EVIDENCE BASE

The technical synthesis that follows depends on two boundaries: which studies entered the review and what each extracted result is allowed to support. This section reports both. It first describes the review design, search, and eligibility rules, then distinguishes the record, report, and study counting units used in selection. The final two parts explain claim governance, technical appraisal, and the evidential limits that shape the survey’s comparisons.

A. Review Design, Search, and Eligibility

We conducted a systematic review with a structured narrative synthesis. PRISMA 2020 organized reporting from identification through inclusion, and PRISMA-S guided the description of search sources, executed queries, dates, and record counts [38], [39]. These frameworks provide a transparent account of the review process; they neither determine technical quality nor constitute the survey’s scientific contribution. The technical scope covers fiber, free space optical (FSO), and VLC/LiFi systems together with eligible terahertz systems assisted by photonics, hybrid systems, and other optical platforms.

The review lineage was registered retrospectively on the Open Science Framework on 12 February 2026 (OSF 7f6wb, DOI: 10.17605/OSF.IO/7F6WB).¹ Searching and screening were already complete, and synthesis and drafting had begun, so this record is not a prospective preregistration. It documents a predecessor state whose last search was conducted on 30 November 2025 and whose denominator was 221 included studies. Rebaselining between June and August 2026 changed the search state, source set, date rule, reviewer process, appraisal framework, and final denominator. The predecessor value of 221 and the final value of 206 must therefore not be read as a sequence in which 15 studies were lost through attrition. Dated protocols, amendments, search logs, decision ledgers, and manifests preserve the two states and their lineage. Electronic Supplement S-Protocol gives each material departure, its rationale, the review stage at which it was identified, and its effect on interpretation.

The primary evidence window ran from 1 January 2020 through 22 June 2026. This start date precedes the earliest eligible report in the final corpus (2021), so the window covers the observed emergence of contemporary O-ISAC under the stated eligibility rules. Searches covered Scopus, IEEE Xplore, ScienceDirect, SpringerLink, Wiley Online Library, and Taylor & Francis Online. Search strings combined ISAC language with optical and photonic terms, including fiber or fibre, FSO, optical wireless, VLC/LiFi, and subterahertz or terahertz systems assisted by photonics. Changes between interfaces were limited to field syntax and query length. Except for the

¹Registration record: <https://doi.org/10.17605/OSF.IO/7F6WB>.

TABLE II
FIELD CONTRACT FOR O-ISAC OBSERVATIONS. CROSS-STUDY STATEMENTS REQUIRE ALIGNED DEFINITIONS, PLANES, TASKS, CONDITIONS, AND PROVENANCE; MISSING FIELDS ARE NOT INFERRED, AND NO UNCONDITIONAL COMPARISON CATEGORY IS DEFINED.

Field group	Required field	Recorded content and allowed state	Failure if absent or misaligned	Required for
Physical context	Modality	Canonical family and active media; controlled family plus segment note	Different propagation regimes appear equivalent	Both
	Target-facing signal path	Ordered source-native path through conversion, target, and receiver	The observable is assigned to the wrong medium or reference point	Both
Functional coupling	Communication task	Payload, control, access, or transport objective; source-native or UNC	Rate or reliability lacks its service meaning	Both
	Sensing task and target	Measurand, event, object, or environment; task–target pair or UNC	Resolution, error, or detection lacks an object	Both
	Integration mechanism	Shared element, resource, path, or rule; controlled multilabel code	An observed relation cannot be attributed to a coupling mechanism	Both
Observation	Metric name and source definition	Label, equation, aggregation, and statistic; source text, equation, or NR	Identical labels can denote different quantities	Both
	Measurement plane	Design, optical, electrical, digital, sensing, or joint/system location	OSNR, SNR, power, or error stages are conflated	Both
	Unit and accounting state	Source-native unit plus gross/net, pre/post-FEC, or summary state	Unit matching hides different payloads, events, or statistics	Both
Operating condition	Operating point	Design variables and resource allocation; condition-set values or NR	Values may occupy different points on a sweep	Both
	Channel, geometry, and environment	Topology, path, distance, motion, and disturbance; condition set or NR	Results from unlike physical regimes are juxtaposed	Both
	Target or ground truth	Source-native target, label, or instrument; NA when inapplicable	Error and detection measures lose their reference	Both
	Constraints	Feasibility, power, bandwidth, timing, or safety limit; equation, statement, or NR	Compared operating regions have different feasible sets	Conditional cross-study
Provenance and decision	Validation type	Analytical, simulation, laboratory, prototype, field, or deployment; controlled code	A bound, simulated estimate, and measurement appear interchangeable	Both
	Baseline or comparator	Source-defined method, configuration, or control; NA if none	Relative improvement has no stable anchor	Both, when comparative
	Value provenance	Reported, source calculated, source digitized, or source derived	Transformation and precision cannot be audited	Both
	Exact locator	Auditable page and section, table, figure, or equation locator	The observation cannot be verified in its context	Both
	Comparison decision and rationale	Decisive alignment or conflict; within-study, conditional cross-study, or descriptive only	The scope of inference remains unspecified	Both

Notes: Requirement states are *required*, *conditionally required*, or *not applicable*. Missing information is recorded as NR, NA, or UNC, never inferred. “Both” means within-study interpretation and a conditional cross-study relation. Curves were not digitized and new performance values were not derived.

two Taylor & Francis exports described below, the log retains the exact string, restrictions, date, export file, and source count for each execution. Date eligibility used the earliest verifiable date of online publication by the publisher or other public availability. Material outside this window could enter only the contextual corpus.

The available source strategies, interfaces, execution log, exported counts, strategy provenance, and execution notes are provided in electronic Supplement S-Search. The exact query to export mapping for two low yield Taylor & Francis exports could not be reconstructed from the audit trail; those rows remain explicitly unmapped and no replacement query was inferred.

Eligible primary reports were peer reviewed journal or early access articles in English and full conference or proceedings papers. The locked protocol required communication and sensing to be considered together through integration, joint evaluation, joint design, or joint optimization. Their common basis could lie in the architecture, optical link, waveform or resource, hardware, channel model, or application setting. A report also had to provide assessable evidence on modality, integration, metrics, tradeoffs, validation, enabling technology, or application. As Section II explains, this boundary extends beyond shared waveform systems. Coupling strength was

consequently characterized after inclusion rather than imposed as a narrower post hoc threshold.

No particular numerical outcome or demonstrated tradeoff was required for eligibility. Explicit 6G wording was coded but was not a keyword only inclusion rule. Weak reporting or limited validation did not exclude an otherwise eligible study; these limitations were handled through claim use restrictions and the technical appraisal described below.

We excluded ISAC studies confined to radio frequency systems with no optical or photonic role, communication without sensing, sensing without communication in the joint system, and reports that did not establish genuine joint consideration. Items available only as abstracts, posters without a full technical report, editorials, opinions, and reports without sufficient English technical content were also ineligible. Reviews and other useful nonprimary records were retained for terminology, historical context, and citation checking, but they did not enter the technical denominator of 206 studies.

B. Selection, Counting Units, and Reviewer Boundary

Records, reports, and studies were treated as distinct counting units. Bibliographic deduplication preceded title and abstract screening, and retrieved full texts were assessed against

the same eligibility criteria. Each exclusion received one primary reason code with a note tied to its source. After eligibility was locked, bibliographic and technical evidence linked reports that described the same experiment, dataset, or system. Journal and conference companions and other related reports remained visible within one study cluster, which prevented report multiplicity from inflating prevalence at study level. Each cluster retained one accessible primary extraction report.

The executed workflow was supervised by the investigator and assisted by AI. Under recorded governance rules, the investigator authorized bounded AI supported adjudication at claim level. OpenAI Codex and Google Gemini supported distinct parts of the recorded workflow [40], [41]. Their roles included record preparation, extraction grounded in PDFs, consistency checks, and proposed adjudication. The provenance trail retained investigator decisions and approvals together with the authorized decision packages. Study progress and method decisions were also discussed and checked verbally in weekly team meetings. This was supervisory oversight rather than independent duplicate screening, extraction, appraisal, or synthesis. Deterministic checks covered identifiers, counts, denominators, source locations, mapping consistency, conflict handling, and workbook invariants. This was not independent double review. Routine independent duplicate human review was not performed, nor were independent human PDF verification of the full corpus and arbitration by a third reviewer. We therefore report no interrater agreement statistic.

One investigator made the final decisions at title and abstract screening, full text eligibility, report to study mapping, extraction adjudication, and appraisal closeout. AI outputs were proposals or structured evidence records, not independent reviewer votes. Eligibility required sufficient English technical content; no translation workflow was used to convert an otherwise ineligible technical body into primary evidence. The versioned forms, codebook, controlled missing-value states, and actual conduct boundary are provided in electronic Supplements S-Data Dictionary and S-Protocol.

Across the six database and platform searches, 1,733 records were identified: 1,273 from Scopus, 329 from IEEE Xplore, 24 from ScienceDirect, 75 from SpringerLink, 29 from Wiley Online Library, and 3 from Taylor & Francis Online. Removing 472 duplicates and 2 records with other metadata or platform dispositions left 1,259 records for title and abstract screening. No record was removed autonomously by an automation tool.

Of these 1,259 records, 927 did not advance to retrieval. This group comprised 864 title and abstract exclusions, 61 records retained only for the contextual corpus, and 2 duplicate or related version flags. The historical retrieval queue therefore contained 332 source records. Consolidating two bibliographic aliases identified after screening produced the formal denominator of 330 unique reports sought. Of these, 58 could not be retrieved, leaving 272 reports for full text eligibility assessment. The preconsolidation retrieval audit contained 60 unresolved source records: 20 links were opened without success, whereas active pursuit of the other 40 was stopped. Thus, the review does not claim that every possible retrieval route was exhausted. Authors were not contacted for retrieval or clarification.

Assessment of the full texts retained 227 eligible primary reports after excluding 39 and assigning 6 to contextual use. The 39 exclusions comprised 4 nonoptical or RF only reports, 2 communication only reports, 12 sensing only reports, and 5 that did not establish genuine sensing and communication integration. Another 4 were abstracts or posters, editorials or opinions, or other reports without a full paper; the remaining 12 lacked sufficient English full technical content. The 61 records retained during screening and 6 retained after full text assessment formed a contextual corpus of 67.

Electronic Supplement S-Exclusions lists all 39 excluded reports with a complete citation and one primary reason. The six contextual reports are retained separately and are not counted as full-text exclusions.

Report to study mapping consolidated the 227 eligible reports into 206 unique studies. The difference of 21 represents journal and conference companions and other reports linked to the same underlying studies, not further full text exclusions. As noted above, this final corpus is not the end of an attrition chain from the retrospective OSF snapshot of 221 studies.

Electronic Supplement ST-01 cites and characterizes all 206 included study clusters and preserves the complete lineage of the 227 eligible reports without counting companion reports as independent studies.

C. Extraction, Appraisal, and Synthesis

Extraction operated at study, report, and claim levels so that technical description remained connected to evidential use. Study records linked bibliographic identity and modality to architecture, integration mechanism, communication and sensing functions, validation, scenario, enabling technology, application, limitations, and reported artifact availability. Claim records preserved the metric, role, value, unit, measurement plane, operating condition, validation context, and an exact page, section, table, figure, or equation locator. A tradeoff entered synthesis only when its affected objectives or resources and an auditable location were identifiable. The schema distinguished reported, calculated, and digitized origins when present, although this review neither digitized graphs nor created new performance values. Missing information was not inferred from nearby text or a companion report.

Extraction targeted source supported results needed to characterize the prespecified survey domains; it was not an exhaustive transcription of every numerical result in every report. When a study reported multiple relevant conditions or result roles, they remained separate metric rows rather than being reduced to a preferred value. Electronic Supplement S-Data Dictionary records field definitions, units, the controlled states NR, NA, and UNC, and the rule that missing information is never inferred.

Study inclusion and claim usability were separate decisions. Claims were assigned to qualitative eligibility, quantitative eligibility, context only use, or quarantined conflict. Quantitative eligibility meant that a value could be used under its recorded conditions; it did not make the value automatically comparable across studies or suitable for pooling. Conditional comparison also required compatible definitions and roles, aligned

units and measurement planes, and sufficiently aligned tasks, scenarios, and validation conditions. Optical OSNR was not substituted for electrical SNR after detection or ESNR without a conversion provided by the source. Bandwidth limited resolution was kept distinct from estimator accuracy and RMSE, as were CRB type bounds and fiber spatial granularity. Gross and line rates were separated from aggregate and net information rates. Analytical, simulation, laboratory, prototype, field, and deployment evidence retained their distinct validation states.

Disagreements between a primary report and its companions were preserved in a conflict register. Values were not averaged, normalized by assumption, or replaced because one appeared more plausible. A quarantined claim was excluded from primary synthesis until an authoritative source resolved the conflict, although unaffected evidence from the same study could remain usable. Context only claims could support terminology or limitations but not primary technical prevalence or performance conclusions.

All included studies were evaluated with the deterministic Technical Quality Assessment Framework (TQAF) developed for this review. Its eight dimensions were each scored from 0 to 3: technical relevance, metric clarity, reporting completeness, validation maturity, reproducibility, benchmark readiness, comparison admissibility, and limitation transparency. A separate field summarized overall evidence contribution. Each score retained its inputs, value before capping, applicable cap, final value, and rationale. An overall score of 3 required technical relevance, metric clarity, reporting completeness, and validation maturity each to be at least 2.

No conventional study level risk of bias instrument was applied. TQAF is a nonvalidated appraisal of technical evidence and reporting specific to this review; it is not a conventional risk of bias assessment and is not GRADE. Electronic Supplement S-Appraisal reports the eight dimension scores, overall contribution category, instrument version, and instrument boundary for all 206 studies. Dimension level rationales remain in the internal calculation audit and are not part of the submitted supplementary materials.

Caps defined by the scoring rules prevented strength in one dimension from hiding a decisive weakness elsewhere. Low metric clarity combined with low validation maturity capped overall contribution at 1; any quarantined claim capped it at 2; and a quarantined metric claim capped comparison admissibility at 1. Limitation transparency was capped at 1 when the authors reported no limitation and at 2 when a material source conflict existed. TQAF characterizes technical evidence and reporting. It is neither a second eligibility screen nor a substitute for reviewer independence.

Study appraisal was aggregated into 115 evidence bodies defined for this review across seven synthesis domains. For every contributing study, the calculation began with the median across all eight TQAF dimensions. For bodies outside S7, certainty was limited when fewer than three studies contributed or the median of the medians calculated at study level was below 1.5. High certainty required at least five studies, a median of the study level medians of at least 2.5, and a median across all eight dimensions of at least 2 in at least 75% of contributing studies. All other bodies outside S7

were moderate. S7 gap bodies used a coverage rule. High certainty required at least ten studies across three modalities, moderate required at least five studies across two modalities, and limited required at least two studies. Remaining bodies were unclear. Broad mixed or unclassified fallbacks were retained for normalization audit, marked nonsubstantive, and excluded from conclusions. These labels are internal synthesis categories, not GRADE certainty ratings.

Heterogeneity was examined descriptively across modality, architecture, metric role, measurement plane, scenario, operating condition, and validation setting. No statistical heterogeneity estimate or meta-regression was performed. Because the review produced no pooled or model-dependent primary effect, a formal sensitivity analysis was not applicable; QA, invariant, and denominator checks were not re-labelled as sensitivity analyses. A formal missing-results or publication-bias tool was also not applied because studies did not share a harmonized prespecified outcome set and source protocols or analysis plans were generally unavailable. Selective outcome and publication bias therefore remain unquantified and cannot be excluded.

Tasks, units, reference planes, operating conditions, and validation designs varied too much to support a common effect size, so no meta-analysis was performed. The synthesis retained the reported conditions across seven domains: modality (S1), integration (S2), metrics (S3), tradeoffs (S4), validation and benchmark readiness (S5), technologies and applications (S6), and evidence gaps and 6G relevance (S7). Mutually exclusive study variables were finalized in a 206-row crosswalk, while multilabel mechanisms, validation types, technologies, and applications were not summed as if they were exclusive. Differences between the executed method and the retrospective OSF snapshot are reported as review limitations.

D. Evidence Base and Inference Limits

Selection determines which studies enter the review; claim governance determines what each extracted item may support. Table III keeps the resulting dispositions, primary record types, and audit events in separate accounting panels.

The reconciliation shows that all 206 studies remain in the survey even when individual claims are restricted: 175 studies had no claim restriction and 31 had at least one. Evidence strength must therefore be assessed at study and evidence-body levels rather than inferred from record volume.

Electronic Supplement S-Evidence supplies the governed primary ledgers behind both panels, including row level study, report, condition, use class, and source locator fields. It contains the 4,779 primary metric rows; the reconciliation records that the broader 4,861-row audit pool also contained 31 context only and 51 quarantined rows, which are neither relabelled nor presented as primary numerical evidence.

TQAF produced complete final scores for 206 of 206 studies. Overall evidence contribution was strong for 125 studies (60.7%), adequate for 75 (36.4%), and low for 6 (2.9%). These review-specific ratings summarize the recorded evidence contribution under the stated rules; they do not revisit eligibility. Technical relevance was strong for 123

TABLE III
GOVERNED CLAIM AND PRIMARY-RECORD RECONCILIATION. COUNTS ARE CODING UNITS, NOT INDEPENDENT EXPERIMENTS OR EFFECTS;
CONFLICT-REGISTER EVENTS ARE EXTERNAL TO THE 8,203-RECORD PRIMARY PARTITION.

<i>Panel A—Claim disposition</i>						
Use class	Records	Unique studies	Conditional subset	In inclusive use set?	In primary synthesis?	Permitted survey use
Eligible quantitative	4,997	206	1,350	Yes	Yes	Numerical or directional evidence under recorded conditions; cross-study use needs a separate gate.
Eligible qualitative	3,206	206	155	Yes	Yes	Source-bounded mechanism, direction, limitation, or qualitative synthesis.
Context only	31	21	27	Yes	No	Terminology, context, or limitations; no primary prevalence or performance claim.
Quarantined conflict	72	31	0	No	No	Withheld pending authoritative resolution; unaffected study evidence may remain usable.
Governed ledger total	8,306	206	1,532	8,234 after quarantine	8,203 after context	Accounting total; use is class specific.
<i>Panel B—Primary record types and external audit events</i>						
Partition	Count	Counting unit	Inside 8,203?	Reconciliation role	Interpretation boundary	
Primary evidence	3,020	Coding record	Yes	Descriptive and qualitative component	Not an independent study or effect.	
Primary metrics	4,779	Metric record	Yes	Typed quantitative component	Definition, plane, condition, and provenance retained.	
Governed tradeoff ledger	404	Ledger row	Yes	Joint relations plus two absence sentinels	Substantive view: 402 records from 168 studies.	
Primary synthesis total	8,203	Coding record	Yes	3,020 + 4,779 + 404	Not studies, experiments, or effects.	
Conflict register	93	Audit event	No	External provenance and resolution trail	Neither record types nor the 72 quarantined claims.	

Notes: $3,206 + 4,997 + 31 + 72 = 8,306$ governed ledger rows; $8,306 - 31 - 72 = 8,203$ primary rows. The primary conditional subset is $1,350 + 155 = 1,505$; adding 27 context-only records gives 1,532 in the inclusive nonquarantined set. Unique-study counts overlap and must not be summed. Coding records are governed and source located; audit events track conflict handling outside the synthesis partition. “Quarantined” means withheld pending authoritative resolution; “conditional” denotes a recorded comparison gate, source guardrail, or condition-dependent decision.

studies, adequate for 68, and low for 15. Metric clarity was strong for 168, adequate for 7, and low for 31. Reporting completeness was strong for 196 and adequate for 10, whereas limitation transparency was strong for 153, adequate for 44, and low for 9. Under the TQAF rules, these reporting and interpretation dimensions were more often strong than the portability dimensions reported next.

The portability dimensions received fewer strong scores. Benchmark readiness was adequate for 158 studies and low for 48, with none in the strongest category. Comparison admissibility was strong for only 4, adequate for 10, and low for 192. Reproducibility was strong for 3 studies, adequate for 199, and low for 4. Validation maturity was strong for 6, adequate for 168, and low for 32. Under these review-specific rules, clear reporting seldom coincided with a reusable benchmark or direct numerical comparison.

Maximum validation setting was recorded separately from the TQAF validation dimension. The former captures the strongest observed setting, whereas the latter also requires evidence for both functions.

The scoring safeguards constrained some final scores. A total of 31 studies with at least one quarantined claim were capped at an overall score of 2, and 26 with a quarantined metric claim were capped at 1 for comparison admissibility. Six studies combining low metric clarity with low validation maturity were capped at an overall score of 1. Separately, 92 metric rows carried forward from the earlier extraction state from 12 studies lacked a final comparability or admissibility label; these were resolved as insufficient information rather

than assigned a favorable status.

Maximum validation tiers, evidence for both functions, and reported artifact availability are interpreted in Section VI.

The study scores populated 115 evidence bodies: 6 for modality, 8 for integration mechanisms, 47 for metrics, 10 for tradeoffs, 3 for validation, 31 for enabling technologies and applications, and 10 for gaps and 6G relevance. Under the review’s internal classification, certainty was high for 54 bodies and moderate for 47; 10 were limited and 4 unclear. The four unclear bodies were broad mixed or unclassified fallbacks retained for normalization provenance, marked nonsubstantive, and excluded from survey conclusions.

Electronic Supplement S-Bodies reports the 115 body ratings and all 4,931 study to body memberships, including the contributing study, synthesis domain, rule used in this review, and certainty label needed to interpret each body.

Together, the study profiles and evidence body classifications define the inference boundary for the remaining survey. Later sections therefore retain reported conditions and admissibility labels, exclude nonsubstantive fallbacks from conclusions, and avoid a universal performance ranking.

IV. OPTICAL PLATFORMS AND INTEGRATION ARCHITECTURES

The evidence base does not describe one optical channel with interchangeable implementations. It describes several physical platforms in which optics can carry the user link, illuminate the target, distribute a wireless carrier, or provide a coherent reference. Those roles determine which impairments

enter the signal path and which design choices can couple communication to sensing. This section therefore moves from physical platform to integration mechanism: first asking where propagation and observation occur, then asking what the two functions share.

For nonduplicative counting, each of the 206 studies was assigned to one canonical modality according to the dominant physical role of optics, whereas integration mechanisms and other attributes remain multilabel.

Canonical modality is therefore a counting rule, not a claim that every system contains only one medium. A fiber-fed photonic transmitter remains in the photonics-assisted terahertz family when the wireless carrier is the principal object of study. It is classified as hybrid only when the interaction between segments is itself consequential to the reported integration.

Table IV fixes the physical counting rule and links each family to the observation path and transfer boundary that the following subsections explain.

The transferable question is where the target perturbs the signal and which element is reused; the numerical answer remains medium and condition specific. The following subsections therefore explain mechanisms rather than repeat the category definitions.

A. Physical Platform Families

1) *Photonics Assisted Terahertz Systems*: In the 69-study photonics-assisted terahertz family, optics enables rather than replaces the wireless chain. Reported architectures include heterodyne or comb-based generation and distribution, photonic delay networks for beam steering, and optical transmitter or receiver processing. Optoelectronic conversion then launches the radiated carrier into a wireless channel. On that path, antenna gain and blockage matter alongside molecular absorption and target scattering; laser linewidth and coherence instead constrain the photonic front end. Modulator response and conversion efficiency matter there as well, but neither describes the target facing path.

Across this family, the recurring design pattern is to use photonics to create wide or mutually coherent signals. Their structure or hardware is then reused for both communication and sensing tasks. Shared multicarrier frames and photonics-assisted phased arrays form one implementation group, whereas other systems use wavelength-tuned chirps or integrate multibeam and LFM-QAM processing [6], [42], [54]–[56]. The coupling may lie in the waveform or reference or instead reside in the beamformer or conversion chain. These designs nevertheless expose the same transfer boundary: optical bandwidth is useful only when phase stability and conversion efficiency preserve it. Its use also depends on aperture and alignment as well as the receiver processing chain.

The observation path reinforces that boundary because a wireless echo may be translated optically for centralized processing, or optical de-chirping may map delay or Doppler to a lower-frequency observable. In either case the target perturbs the radiated field, whereas source and front-end drift enter

through a different path. Laboratory studies can isolate waveform or beamforming behavior; outdoor demonstrations additionally test propagation and continuity. Comparison within this family must therefore cover the complete optical-to-wireless chain and its validation setting, not carrier frequency or headline rate alone.

2) *Fiber O-ISAC*: The 56 fiber studies use the guided path both to transport information and to observe a disturbance. Intrinsic backscatter and reflection provide one class of sensing observable. Other systems rely on polarization evolution or phase, on loss, or on a deliberately introduced element such as a grating. The relevant spatial coordinate follows the fiber route rather than a free-space geometry. Consequently, the meaning of a reported sensing granularity remains tied to the interrogation method and topology.

Integration spans two broad regimes. Some systems reuse communication pilots or subcarriers, including deliberately joint signals. Others introduce a probe or wavelength while sharing the deployed fiber and either the source or receiver. Digital subcarrier coexistence and co-wavelength distributed acoustic sensing illustrate the first regime, and joint PAM/LFM signaling extends it. Submarine cable monitoring and live network OTDR broaden the deployment setting [1], [36], [43], [57], [58]. Sharing the cable differs from sharing a waveform, yet it can still couple launch power to wavelength occupancy and receiver headroom.

In the reviewed fiber evidence, dynamic range and reference stability recur as interpretive constraints because weak returns must be recovered beside leakage and reflections, often amid noise and strong data-bearing components. Additional launch power can help the return while aggravating non-linearity or saturation elsewhere, whereas coherent sensing preserves phase but requires a stable reference and traffic-aware cancellation. Evidence also changes with deployment: a spool experiment isolates interaction, an access network tests coexistence with splitters and traffic, and an operational route tests environmental drift and service continuity. These settings answer different questions even when they use the same nominal fiber length or sensing label.

3) *VLC/LiFi Systems*: The 38-study VLC/LiFi category supports consistent review-level counting; it does not make visible-light communication and LiFi synonymous. The evidence spans point links and networked access, with some systems using luminaires or screens alongside cameras and photodiodes. Others use vehicle lamps or wearable devices. Designs based on intensity modulation and direct detection must account for non-negativity and source linearity, while dimming and flicker can enter the same design problem. Field of view and ambient light matter as well. Cameras add spatial information but impose exposure and frame-rate constraints; photodiodes support faster signals with less intrinsic spatial discrimination.

The shared physical pattern is that emitted or reflected light carries both payload and task information, but the task changes what that information means. Some systems combine control data with robot pose or vehicle packets with distance and maneuver sensing. Others couple VLC to gesture recognition or to a closed reflected-light control loop, while wearable

TABLE IV

CANONICAL PHYSICAL ORGANIZATION OF THE 206-STUDY CORPUS. STUDY COUNTS ARE EXCLUSIVE; MEDIA AND INTEGRATION MECHANISMS MAY OVERLAP. COUNTS INDICATE EVIDENCE ACCUMULATION, NOT PERFORMANCE OR MATURITY.

Family and studies	Optical role and target-facing path	Sensing observables or tasks	Typical coupling	Active constraints	Numerical transfer boundary	Evidence anchors
Photonics assisted THz, 69/206 (33.5%)	Optically generate, distribute, or process a coherent carrier; target path is wireless THz or millimeter wave after conversion	Delay, range, Doppler, angle, imaging, channel or target response	Hardware, reference, waveform, beam, resources	Linewidth, coherence, conversion, aperture, alignment, absorption	Align the optical-to-wireless chain, aperture, receiver processing, and validation	[6], [7], [42]
Fiber, 56/206 (27.2%)	Guided traffic path exposing backscatter, reflection, phase, polarization, or loss; disturbance acts along fiber or an inserted element	Distributed event localization, vibration, strain, environment	Hardware, wavelength, waveform, link, resources	Dynamic range, leakage, reflection, nonlinearity, reference, topology, traffic	Match interrogation, topology, span, launch, receiver, traffic, and validation	[1], [36], [43]
VLC/LiFi, 38/206 (18.4%)	Illumination or access through emitters, cameras, and photodiodes; direct or reflected room/device path	Position, pose, distance, imaging, gesture, recognition, physiology	Hardware, waveform, geometry, resources, application loop	Nonnegativity, dimming, flicker, linearity, ambient light, field of view, exposure	Match task, layout, orientation, illumination, detector, and processing	[2], [44], [45]
Free space optical, 31/206 (15.0%)	Directional link and optional target illumination/return; direct or reflected aperture and beam path	Ranging, localization, imaging, tracking, atmosphere or channel	Link, waveform, aperture/beam, spatial and temporal resources	Pointing, divergence, aperture, reflectivity, loss, turbulence, motion	Match geometry, calibration, target, weather, motion, and validation	[46]–[48]
Hybrid optical, 9/206 (4.4%)	Consequential optical, fiber, free space, or radio segments; perturbation and observation may straddle conversion	Segment-specific range, vibration, target, or network state	Central reference; cross-segment link, resources, control, application	Conversion loss, segment power, synchronization, calibration, timing	Trace every segment, conversion, clock, plane, and function-specific validation	[34], [49], [50]
Other optical, 3/206 (1.5%)	Source-specific optical control or coherent carrier; microwave field, solar-blind UV, or quantum optical path	Field pattern, particulate/smoke response, or phase	Source-specific hardware, reference, and application	Incompatible propagation, target, receiver, and measurement models	Case-specific interpretation; no numerical transfer across the residual category	[51]–[53]

Notes: Canonical modality is mutually exclusive for counting; physical components and integration mechanisms are not. Hybrid requires consequential interaction between segments. Photonics assisted THz retains a radio-frequency target path. Coupling entries are recurring, not universal, family properties.

devices add communication with physiological sensing [35], [44], [45], [59], [60]. These systems may share a source or receiver and sometimes their geometry or application loop, even though they need not share the same waveform or observable.

In the reviewed configurations, illumination creates the family's distinctive transfer boundary. Increasing modulation depth or receiver field of view can benefit one outcome while reducing optical headroom or admitting background and multipath. Bias and clipping also change the residual signal from which motion is inferred. Localization depends on geometry; gesture and recognition tasks depend on reflection contrast; and health applications also depend on model generalization. Results can therefore be compared within a matched task and setup, not through a common VLC/LiFi sensing score.

4) *Free Space Optical Systems:* The 31 FSO studies place the optical carrier in a directional free-space path. Evidence comes from direct and retroreflective links as well as beam scanning or steering. These systems combine data delivery with ranging or localization; others add imaging, tracking, or channel observation. Unlike the illumination-centered VLC/LiFi family, their designs are organized chiefly around aperture coupling and beam control. Beam divergence and pointing shape both functions, while receiver aperture and target reflectivity add another dependency alongside atmospheric loss and turbulence.

The experiments illustrate several ways of using that geome-

try: multicarrier signals can support data and range processing. A steered source can then sequence localization before imaging and communication, while multiple access points infer position and co-path turbulence observations inform beam shaping [3], [46]–[48], [61]. These arrangements do not form a common performance scale. Narrow beams improve spatial selectivity but tighten acquisition and tracking, whereas wider beams or fields of view ease alignment while reducing power density or admitting background.

In these FSO studies, the direct data path and reflected sensing path retain different timing and power budgets. A common clock or waveform can relate the estimates without making their geometries identical. Consequently, absolute time of flight and relative delay require different calibration, as do angle estimates and image-derived features tied to received-power geometry. Controlled links establish waveform or estimator behavior, whereas outdoor and operational tests add atmospheric variation and motion while also testing continuity. Findings transfer only when those geometric and validation conditions align.

5) *Hybrid and Residual Optical Systems:* Nine studies are hybrid because more than one physical segment is consequential to the integrated result. Fiber distribution may feed a wireless sensing path while fiber and radio front ends share control; other services coordinate THz with VLC or FSO. The evidence includes radio-over-fiber ISAC and fronthaul pilots reused for distributed vibration sensing. Heterogeneous

photonic transmitters form another case alongside coordinated THz-VLC access and FSO paired with fiber-grating sensors [34], [49], [50], [62], [63].

Across the reviewed hybrid cases, the analytical value lies at the interfaces. Optical power in a fiber span differs from RF power at an antenna, while received optical power at an FSO terminal occupies another segment. Delay and synchronization may span central processing and transport before conversion and propagation. Even so, centralization can improve coherence or reuse while introducing conversion loss, calibration burden, or a new timing bottleneck. Hybrid claims therefore require an end to end account of where information enters and where it is converted, together with what is shared and where it is observed.

Three further studies remain outside the five recurring families: optical microcombs used to control structured microwave fields, solar-blind ultraviolet communication combined with particulate or smoke sensing, and coherent-state quantum optical communication with phase sensing [51]–[53]. The residual category preserves eligibility without manufacturing a common propagation or receiver model, so its evidence is interpreted case by case rather than pooled.

Taken together, the family-specific mechanisms and constraints motivate the architecture analysis that follows.

B. Integration Architectures and Resource Coupling

The modality families establish where signals propagate; integration mechanisms establish what the two functions actually share. This distinction matters because identical hardware can carry separate signals, while a shared waveform can be processed by separate receivers. Building on the coupling locations defined in Section II, this section examines the observed coupling mechanisms rather than treating integration as a maturity label.

The coding is multilabel because several mechanisms often coexist. Shared resource allocation was identified in 118 studies, shared hardware in 117, and a shared waveform in 113. Shared link or channel use appeared in 87 studies, joint design or optimization in 72, shared optical carrier use in 49, and a shared application scenario in 46. Three studies required an additional mixed label, and none remained unresolved after normalization. These totals must not be added or read as competing prevalence rankings.

1) *Shared Hardware, Carrier, and Link*: Physical reuse may couple functions through a source and modulator or through a photonic circuit. The shared element may instead be an aperture or path, and a receiver or reference can play the same role. Such reuse can reduce duplication but may tie operating points to linewidth and bandwidth or to dynamic range and alignment. The informative unit is therefore not “shared hardware” alone but the component, its role in each function, and the constraint transferred between them.

The evidence shows how that rule changes with the signal path: in fiber, a common waveguide can carry traffic while reflections and phase changes reveal events. Polarization and distributed scattering provide other observables, so integration may reside in the optical structure or transceiver or in both

[1], [64]. In FSO, a directed or retroreflective path can couple information delivery to distance or position inference [46]. Photonics-assisted arrays may instead share optical generation and beam-control hardware while the target-facing path remains wireless [6]. The same hardware label consequently carries different propagation and calibration implications.

Carrier sharing is a narrower claim because it requires the same wavelength, optical carrier, or carrier derived signal to participate in both functions; two wavelengths in one chassis may share hardware without sharing a carrier. A common carrier may also feed branches with different receivers. This distinction matters when coherence, phase relation, or wavelength reuse is offered as the benefit, because those benefits do not follow from generic hardware colocation.

2) *Shared Waveforms*: Waveform sharing places the coupling in signal structure, where data symbols and pilots may support both decoding and a sensing estimate. Chirps and codes can serve the same purpose, as can polarization states or spatial modes. The functions may use the entire signal or only structured parts of it. The defining feature is technical relevance to both tasks, not mere simultaneous transmission.

The studies reveal several variants of this mechanism: pilots inserted for channel estimation can also supply a sensing reference, directly linking pilot support to payload efficiency and power distribution [65]. Communication symbols can instead be embedded in a ranging waveform, as in a photonics-generated amplitude–frequency-modulated terahertz signal [66]. Visible-light systems can couple data to spatial, reflected, or camera-observed structure within an adaptive control loop [2]. These examples share a design principle but not one modulation or receiver model.

Waveform sharing also does not establish concurrency by itself. The coding did not infer concurrency when a study alternated segments, estimated sensing quantities offline, or evaluated the two paths under partly different conditions. Coding the mechanism records a technical relationship; whether both outcomes were measured simultaneously at a matched operating point remains a separate evidence question addressed in Section V.

3) *Resource Allocation and Joint Design*: Resource coupling extends beyond waveform reuse because time and bandwidth may be divided, as may wavelength and subcarriers. Other systems coordinate optical power or polarization, while spatial beams and receiver headroom or processing capacity create further limits. Each resource creates a different candidate coupling mechanism: pilot allocation changes payload opportunity, while power allocation changes margin and detectability. Wavelength separation may suppress interference while consuming spectrum otherwise available to payload; beam allocation changes geometry and coverage. A generic “ISAC resource” would hide those mechanisms, so the resource and allocation rule must remain explicit, together with the constraint and objective.

Joint design is present when communication and sensing enter the same rule or objective, and they may also share a constraint set or adaptive policy. A visible-light V2X camera system links transmission and sensing through its resource decisions and geometry [67]; a fast-swept photonics-assisted

terahertz system couples radar timing and data transmission through a common source and waveform plan [7]. In both cases, “joint” identifies coordinated design variables, not a guarantee that the solution is globally optimal or portable.

That distinction separates architecture from outcome. A weighted model, a configuration sweep, and one feasible operating point provide different forms of evidence about coupling. None alone establishes a transferable Pareto frontier or deployment policy. Section V therefore synthesizes observed relationships separately from these mechanism labels.

4) *Application Level Coupling and Boundary Cases:* At the broad edge of the review boundary, sensing and communication may meet through beam selection or handover. The connection may instead lie in trajectory planning and blockage response or in security and service control. None of these cases requires a common signal. Retaining a shared-application category keeps such systems visible without silently upgrading them to shared-waveform integration. A near-infrared device can join the functions through hardware and use context [35], whereas an FSO–fiber system may coordinate separate links for one network service [50]. Their coupling locations are materially different.

The three mixed cases make this boundary explicit. They include sequential target recognition and mixed FSO–RF network control [61], [68], [69]. The label is not a quality penalty; it records a design that crosses the controlled categories or segments. Subsequent interpretation must therefore identify the segment and function to which each result belongs.

C. Cross-Platform Interpretation

Read together, the platform and architecture views explain why O-ISAC cannot be organized as a simple catalogue of technologies. A shared waveform in fiber acts through guided propagation, reflections, and receiver headroom; a shared waveform in an FSO or terahertz system also inherits aperture, geometry, and alignment. Conversely, systems in one platform may range from a shared link with separate signals to a common waveform governed by one allocation rule. The platform locates the physics, while the architecture identifies the design choice that transfers a constraint between functions.

Sharing additional elements can reduce duplication or improve coherence, yet it can also increase interference, synchronization burden, dynamic range pressure, or failure coupling. The taxonomy is therefore descriptive rather than hierarchical. Its analytical output is a mechanistic question: which reported outcomes changed with a shared choice, and under what operating conditions? Section V addresses that question by defining the measurement and comparison contract before synthesizing the observed joint design relationships.

V. PERFORMANCE METRICS AND JOINT DESIGN TRADEOFFS

The architectures in Section IV identify what communication and sensing share, but an architecture label does not reveal the effect of that sharing. The effect becomes evidence only after the reported quantity, reference point, operating conditions, and validation setting are known. This

section therefore follows an analytical sequence: it defines the measurement contract, interprets communication and sensing outcomes within that contract, and then traces reported relationships associated with shared design choices.

The primary synthesis contains 4,779 metric records, which are typed reported quantities rather than independent effects. Study coverage is multilabel, and record volume reflects extraction granularity; neither count ranks scientific importance or platform maturity.

The most widely covered named families are data rate in 85 studies, bit error rate (BER) in 60, and range resolution in 50. Coverage then fragments across link and waveform quantities as well as device and estimator outcomes, with application and implementation measures adding further variation. The residual “other” code is a governance bucket rather than a performance family. The synthesis question is therefore not which label occurs most often, but which observations are aligned closely enough to answer a stated engineering question.

A. Measurement and Comparison Contract

We interpret each metric as a typed observation whose source definition and task establish its semantics. The measurement plane and scenario then locate the observation, while the operating point and validation method bound it. A scalar rate remains incomplete without its payload accounting, just as BER needs a receiver stage and coding state. A sensing error needs its measurand and ground truth, together with the estimator and summary statistic. Matching names without these fields creates agreement in notation rather than in evidence, which is why experimental settings remain attached. Bandwidth and carrier may be inputs rather than outcomes, as may subcarrier count or launch power. Target distance and hardware settings complete the configuration that bounds each reported value. Two results in one paper form a joint operating point only when their condition sets show that they were acquired together. Provenance further distinguishes a stated threshold from a tabulated value and an explicitly labeled figure value from a value calculated by the source. We did not digitize curves or derive new performance values, so narrative synthesis does not manufacture precision that the sources did not report.

Measurement planes make these distinctions operational. Configured bandwidth and pilot fraction belong to the design and input plane, as do launch power and geometry. Received optical power and OSNR belong to the optical link. After conversion, electrical SNR and equalizer outputs inherit the detector and sampling chain. EVM, BER, and FEC status further depend on the reference bandwidth and DSP state. A value expressed in decibels cannot cross these planes merely through unit matching.

The sensing plane links an estimate or decision to its target and ground truth, whereas the joint and system plane contains matched tuples and end to end outcomes. Multisegment systems make this distinction especially important. A fast swept photonics assisted terahertz architecture contains optical source and device records alongside wireless link and system

evidence [7]. No record alone describes the complete chain. Plane assignment preserves useful connections without treating inputs as link conditions or receiver outcomes as sensing decisions.

The evidence contains many familiar performance labels, but their frequency does not establish common meaning. Table V reconciles metric-domain coverage and converts those labels into explicit comparison requirements.

The low conditional-admissibility share follows from heterogeneous definitions and conditions; it is not a judgment that the remaining records are uninformative. The function-specific synthesis therefore uses those records for mechanism, feasibility, and within-study interpretation while withholding unsupported cross-study arithmetic.

B. Metrics Across Functions

1) *Communication Metrics*: Communication results first separate signaling pace, information accounting, delivered service, and theoretical limits. Symbol rate describes signaling pace. Gross and net data rates depend on payload and overhead accounting, whereas aggregate and bidirectional values extend the scope of that accounting. Throughput describes delivered data under stated conditions, capacity is a model dependent bound, and spectral efficiency normalizes a defined rate by bandwidth. One photonics assisted terahertz prototype reports net rate with its sensing and communication subcarrier partition, making the frame definition part of the result [42]. The reported efficiency in the embedded-pilot OFDM study is likewise the fraction of data bearing subcarriers, but its definition excludes guard intervals, cyclic prefixes, and other implementation losses elsewhere in the signal chain [65].

Reliability metrics locate different error events because pre-forward-error-correction (pre-FEC) BER differs from post-FEC BER. Packet loss and outage probability describe other events, while error vector magnitude and optical Q-factor probe signal quality. These measures cannot substitute for one another. A long-span fiber experiment reports optical signal-to-noise ratio (OSNR) and pre-FEC BER beside its sensing outcomes [70]. A robotics experiment uses packet loss across distance and view angle and then observes its effect on display delay and control [44]. Both address reliability, although their events occur on different time scales and across different system boundaries.

Signal-quality and power references further constrain interpretation because OSNR is defined in the optical domain and depends on a noise bandwidth. Electrical SNR after photodetection and digital SNR after equalization include different transfer functions, while launch and received powers refer to different points in the link. A fiber polarization-sensing system reports data rate and OSNR alongside sensing SNR, as well as estimation and localization errors [32]. Their coexistence provides system context; it does not place the values on one numerical scale.

Distance and latency are similarly conditional because fiber span and wireless separation describe different geometries, as do target range and coverage radius. Latency may cover propagation or buffering, or it may instead include acquisition

and processing or extend through inference to the application loop. We therefore use these quantities to describe operating regimes and bottlenecks, not to rank modalities by reach, speed, or reliability.

2) *Sensing Metrics*: Sensing interpretation begins with the task because ranging and localization do not share one accuracy scale with velocity or vibration. Imaging and classification introduce different outcomes, as does environmental measurement, while resolution and estimation error ask different questions even within one task. The former concerns separability, whereas the latter compares an estimate with a reference. Repeatability and a model-derived lower bound add still different evidence, so a valid comparison must retain the source's operational definition rather than normalize these distinctions away.

The studies make this boundary concrete. An OFDR system reports breakpoint localization error and range resolution under experimental and simulated conditions [37]. A visible light localization system separates lateral error from demonstrated separation and reports signed displacement with trial level variation [2]; the reported summary matters as much as the outcome. Mean absolute error and root mean square error emphasize different properties of a distribution. Median and maximum values reveal other behavior, whereas a confidence interval or percentile quantifies uncertainty. Thresholds and curves answer different questions again. An optical phased array result tied to simulated electrical SNR and beamforming conditions is informative within that model, not a platform independent ranging score [33].

Detection and classification add dataset and decision context because accuracy and precision depend on class balance and the decision threshold. Recall and false alarm probability also depend on the decision threshold and background conditions; learned systems additionally require a documented separation between training and test data. CORE-Lens combines recognition accuracy with camera-communication BER and runtime [71]; VehicleTalk reports link outcomes alongside distance and vehicle-status recognition in a prototype [45]. These are joint application results, but their class definitions and geometries preclude a direct accuracy ranking. Physical-quantity sensing similarly requires a defined measurand and reference instrument, while calibration and spatial support must also be known, together with acquisition time and the aggregation rule, before a value can support numerical comparison.

3) *Joint and Implementation Metrics*: In this synthesis, a joint result is treated as a matched tuple when its condition linkage is documented, rather than as a special ISAC score. The embedded-pilot example combines BER, pilot-dependent efficiency, and a source-defined mean absolute pilot error [65]. That last quantity is a pilot-deviation proxy in the reported simulation, not a universal environmental-sensing error. The tuple is useful because a common allocation links its components, not because they collapse to one scale.

Optimization results have the same boundary. Communication constraints and sensing criteria may determine one operating point. Power and geometry also matter, as does the weighting used in optical phased-array beam pattern design [33]. Changing a constraint or normalization changes the

solution. A weighted objective demonstrates co-design under a model; it does not establish a universal utility function or transferable frontier.

Implementation metrics show whether that design survives a platform. Receiver bandwidth and converter rate may become the active constraint. Memory and latency create a different bottleneck, while acquisition time can expose power or calibration burdens. An outdoor photonics-assisted terahertz study reports receiver-pipeline latency with communication and ranging outcomes [42]. Downstream application metrics absorb still more choices. Robot behavior depends on link and display delay [44], while camera-based recognition combines BER with runtime and joint-cycle latency [71]. These outcomes can be decision-relevant without becoming intrinsic properties of a modality.

C. Recurring Tradeoff Mechanisms

The governed tradeoff ledger records any evidenced coupling created by a shared choice or constraint. Some couplings are antagonistic, such as pilots displacing payload, whereas others show joint improvement until a bottleneck changes. We therefore distinguish strict tradeoffs from conditional or synergistic relationships rather than treating every ledger row as a loss on one side.

The governed ledger contains 404 coded rows linked to 169 studies. Two rows record the absence of a reported tradeoff: one accompanies substantive records from the same study, and the other is the only row for its study. Removing these audit sentinels leaves 402 substantive records from 168 studies: 218 eligible for quantitative use and 184 for qualitative use. Each row captures a coded relationship rather than a separate experiment or effect estimate. A study may contribute several records or appear in several mechanism families, so family study counts must not be added and frequency does not measure effect strength.

Conditionality is the defining feature of the evidence. Of the 402 records, 371 (92.3 percent) carry a comparison gate, source guardrail, or condition-dependent admissibility decision. A quantitative record preserves a numerical relationship within its experiment or model; it does not make the relationship portable across waveforms, targets, geometries, measurement planes, or validation settings. The families below synthesize recurring reported coupling mechanisms while retaining that boundary.

1) *Bandwidth, Spectrum, and Resource Allocation*: Bandwidth, spectrum, and resource allocation form the largest family: 94 records from 70 studies, comprising 50 quantitative and 44 qualitative records. Eighty-nine records (94.7 percent) are conditional. The shared constraint appears as subcarrier occupancy, wavelength use, guard intervals, acquisition time, or optical and electrical bandwidth. These resources all limit signal dimensions, but they act at different planes and cannot be reduced to one overhead variable.

The representative evidence shows why the distinction matters. More embedded OFDM pilots reduce the source-defined pilot-deviation error while increasing BER and reducing the data-bearing fraction. In co-wavelength fiber integration, a

wider protection interval improves sensing isolation but encounters communication-device bandwidth limits. In a fronthaul prototype, a wider LFM pilot sweep improves pulse-compression resolution while increasing overlap and BER [34], [43], [65]. Pilot support, spectral isolation, and chirp bandwidth thus move outcomes through different mechanisms even when each reduces some notion of payload opportunity.

2) *Power, Energy, and Dynamic Range*: Power, energy, and dynamic range contribute 90 records from 64 studies: 50 quantitative, 40 qualitative, and 81 conditional (90.0 percent). Their common feature is a bottleneck whose meaning depends on the reference plane. Launch power, received optical power, payload-to-probe balance, carrier-to-signal ratio, device bias, modulation swing, and converter headroom describe different controls. Increasing one can improve a weak return while worsening nonlinearity, saturation, clipping, or the margin available to data symbols.

A fiber LFM-probe experiment illustrates direct competition: increasing the sensing-to-telecom power ratio improves sensing SNR but worsens BER, especially near the probe. A visible-light prototype instead splits optical power among communication, ranging, and energy harvesting; more power to the first branch raises rate and lowers ranging error while reducing harvested power. Its two functions were also measured separately because of prototype constraints [72], [73]. These examples support a common rule: power allocation is interpretable only with its reference point and active bottleneck. A region of joint improvement can indicate that the active bottleneck has shifted; it does not show that coupling has disappeared.

3) *Communication Reliability and Sensing Quality*: The reliability and sensing family contains 59 records from 48 studies: 37 quantitative, 22 qualitative, and 55 conditional (93.2 percent). Here the coupling often appears after the waveform is fixed. The coded explanations for changes in BER, EVM, Q-factor, required OSNR, sensing SNR, estimation error, or detection measures include nonlinear mixing, leakage, imperfect cancellation, phase noise, and receiver competition.

Two studies show that avoidable interference should be separated from an unavoidable allocation loss. Sequence shaping in co-directional communication and CC-OTDR reduces probe-induced errors while retaining distributed sensing. A photonics-aided millimeter-wave design instead uses DC offset to improve sensing SIR at the cost of communication EVM [74], [75]. The first changes sequence transitions and fiber interaction; the second changes carrier energy and constellation displacement. Together they suggest a staged design rule: mitigate avoidable coupling before treating an observed loss as an inherent allocation tradeoff. They do not support a common numerical exchange rate.

4) *Rate, Resolution, Accuracy, and Range*: Rate and resolution form the fourth largest substantive family in this corpus: 46 records from 39 studies, or 11.4 percent of the 402 substantive records. The family contains 26 quantitative, 20 qualitative, and 41 conditional records (89.1 percent). A direct relationship arises when bandwidth, symbol duration, coding, or frame allocation determines both information opportunity and target separability. An optoelectronic-oscillator design

links single-channel capacity to maximum range, whereas remote phase-shift LiDAR changes the relation between bit rate, nominal distance resolution, and unambiguous range [76], [77]. Both relationships are design-specific; neither establishes that higher data rate universally implies poorer range resolution.

The broader relation between rate and accuracy or localization contains 40 records from 33 studies: 22 quantitative, 18 qualitative, and 39 conditional (97.5 percent). Its breadth is substantive rather than terminological. A retroreflective system varies the power ratio between ranging and OFDM, moving ranging error, BER, and FEC-limited reach. An optical-camera system improves communication-stripe recovery at the cost of background reconstruction and recognition quality [78], [79]. The controls and sensed objects differ, so the common family supports mechanism comparison rather than numerical pooling.

The relation between rate and range or coverage contributes 21 records from 16 studies: 10 quantitative, 11 qualitative, and 18 conditional (85.7 percent). Its coupling is usually temporal or geometric. Pulse repetition and observation duration exchange update opportunity for unambiguous range, while beamwidth exchanges link margin or angular selectivity for coverage. Index-modulated LFM and time-division W-band studies make these controls explicit [80], [81]. Their operating tables remain conditional on frame, interference, and propagation assumptions rather than forming a cross-platform frontier.

5) *Waveform, Hardware, and Complexity*: Waveform, hardware, and complexity account for 21 records from 20 studies. Only 6 are quantitative, 15 are qualitative, and all 21 are conditional. The qualitative emphasis reflects how complexity is argued in these records: component reuse, coherence, or real-time feasibility is proposed without a matched performance-versus-cost curve. A design may reduce source or receiver count while transferring burden to calibration, synchronization, or digital processing.

A full-photon THz study contrasts simple free-running lasers with a more complex coherent-comb chain needed to control phase noise and drift. A fiber experiment takes a different route, using a broad-linewidth low-cost source and residual-carrier processing to recover vibration [82], [83]. These systems cannot be ranked by block count alone. Physical components, calibration and synchronization, computation, memory, and latency describe separate dimensions of implementation burden.

6) *Implementation and Boundary Relationships*: The remaining codes retain relationships that would be distorted by the recurring families. Other joint tradeoffs contribute 19 records from 15 studies: 14 quantitative, 5 qualitative, and 18 conditional (94.7 percent). Qualitative or partial general relationships add 10 records from 8 studies: 2 quantitative, 8 qualitative, and 7 conditional (70.0 percent). Security or resilience and synergistic or non-antagonistic coupling each contain one conditional record from one study; the former is quantitative and the latter qualitative. These are boundary codes, not mature evidence families.

The synergy case is nevertheless conceptually important.

Communication and sensing can improve together when a common reference, cancellation method, or channel estimate benefits both, with antagonism emerging only when bandwidth, power, dynamic range, latency, or hardware becomes binding. The location and shape of the relationship are therefore properties of a specified system and condition set. A single security or synergy record cannot support a general frontier.

D. Limits of Cross Study Comparison

The final admissibility decision in Table V is deliberately stricter than matching metric names. There is no unconditional *yes* category, and a conditionally admissible record remains tied to its source definition and measurement plane as well as its task, scenario, operating conditions, and validation type.

A conditional decision permits a narrow comparison only after every guardrail is satisfied. It can align the same BER stage and modulation under matched link conditions, or the same localization error definition and ground truth protocol. It cannot combine a threshold with a mean, a simulated bound with a measured error, or rates with different payload accounting. Unit conversion does not repair semantic or experimental mismatch.

A “no” decision does not make a result uninformative. Such records can explain a mechanism, establish feasibility, preserve a within study baseline, or support qualitative tradeoff synthesis. They are withheld from cross study arithmetic when decisive definitions or conditions differ. Accordingly, this survey neither pools metric values nor ranks modalities by reported maxima.

The same boundary applies to the 402 substantive tradeoff records. A Pareto curve is interpretable only within the objectives, constraints, metric definitions, and measurement planes that generated it. Some records contain parameter sweeps or selected operating points; others compare architectures or state only a directional relationship. They do not sample one shared design space and therefore cannot support a universal rate versus resolution law or a universal performance frontier. They should instead be read as a map of reported coupling mechanisms. The two absence status rows remain in the governed ledger solely to preserve audit coverage.

This comparison discipline also fixes the next evidential question. Even a well defined matched tuple has limited engineering value if it appears only in an opaque simulation or an incompletely specified prototype. Section VI therefore examines how the reported relationships were validated, whether the two functions were tested together, and whether the configuration can be reconstructed for benchmarking.

VI. VALIDATION, REPRODUCIBILITY, AND BENCHMARK READINESS

An experiment label alone does not establish validation maturity in O-ISAC. Simulation can isolate impairments that a prototype cannot control, whereas a laboratory system can expose hardware effects without testing mobility or environmental variation. Even field evidence may exercise only one of the two functions. We therefore distinguish the strongest

setting reached by a study, the validation methods it used, and whether communication and sensing were validated together. Reproducibility adds a separate requirement: an external reader must be able to reconstruct the configuration and evaluation contract.

A. Maximum Validation Maturity

The mutually exclusive maturity distribution places 32 studies (15.5%) at basic simulation or numerical evidence and 18 (8.7%) at enhanced simulation or dataset-supported validation. Higher tiers contain 78 (37.9%) laboratory experiments or proofs of concept, 66 (32.0%) controlled prototypes, and 12 (5.8%) field trials or deployments. These counts sum to 206 because each study enters only its highest observed tier. They describe the strongest setting reached, not every method used along the way.

A total of 156 studies reached at least a laboratory or proof-of-concept tier, but only 12 reached field trial or deployment. The corpus therefore contains substantial evidence beyond analytical feasibility, although operational evidence remains limited. Simulation-led examples isolate waveform and device sensitivities [65], [84]; laboratory and prototype systems expose conversion, synchronization, and receiver effects [35], [49], [85]. Field-oriented studies extend the evidence to access networks, deployed fiber, submarine monitoring, or environmentally exposed wireless links [36], [42], [86]–[88]. These examples distinguish validation settings rather than rank performance.

The maximum tier does not measure completeness within a setting. A field trial may cover one route or disturbance, while a controlled prototype may explore a broader parameter space. Conversely, a common-clock laboratory experiment that measures both functions can reveal their coupling more directly than a field dataset that records only one outcome. Maturity must therefore be read together with coverage of conditions, controls, and functions.

B. Validation Methods and Operating Realism

The overlapping validation map explains how studies build that evidence. Analytical treatment was recorded for 131 studies and numerical analysis for 14; simulation appeared in 104 and dataset-based evaluation in 13. Laboratory experiments appeared in 148 studies, prototype or testbed validation in 83, field experiments in 12, and mixed validation in 33; no study remained unclear. The totals exceed 206 because a study can connect analysis and simulation to measurements and a prototype. For the same reason, 148 studies can report laboratory evidence although only 78 have laboratory work as their maximum tier, and 83 can report a prototype although only 66 have that maximum.

These methods answer different validity questions. Analysis states assumptions, simulation supports controlled sweeps, and laboratory work exposes real components. A prototype tests interactions among subsystems; a field experiment tests at least one operational boundary. Combining methods is useful only when their configurations and transitions are traceable.

A mixed label is therefore not an automatic mark of greater rigor.

Operating realism is also modality-specific. Fiber validation may require live traffic, deployed topology, and environmental events. FSO and photonics-assisted terahertz systems may require pointing and atmospheric exposure, while VLC/LiFi adds ambient light, orientation, motion, and illumination constraints. A useful validation claim must identify the propagation environment, measurement plane, target or disturbance, reference method, and the function that was actually exercised.

C. Maximum Tier Versus Paired-Function Evidence

The review-specific TQAF validation dimension imposes a stricter condition. Its distribution is 32 studies at score 1, 168 at score 2, and only 6 at score 3. The strongest score requires field or deployment evidence for both communication and sensing. Thus, all 12 field/deployment studies occupy the highest maximum tier, but only half satisfy the paired-function rule. The other six retain valuable field evidence without establishing both functions at that setting.

The study-level crosswalk is supplied as Supplement S7 in `s7/S7_PAISED_FUNCTION_VALIDATION_12.csv` and `s7/S7_PAISED_FUNCTION_VALIDATION_12.xlsx`. It retains all 12 maximum-tier records, marks the six strict paired-function cases, and leaves unresolved function-specific source locators explicit rather than treating them as matched evidence.

This rule prevents evidence from migrating silently between domains. A deployed sensing record does not validate communication under the same conditions, just as a live link and a laboratory-only sensing result do not jointly constitute a field trial. Field-collected fiber data with offline classification illustrates the distinction [89]. In segmented architectures, evidence continuity additionally requires traceable interfaces, timing, calibration, and concurrent operating conditions. Mature components tested separately do not by themselves establish a mature end-to-end operating point.

D. Reported Artifacts and Reconstructability

Artifact reporting reveals a different bottleneck. Artifact availability and reconstructability are related but not equivalent. Table VI separates the reported access route from the information required to rerun the full system chain.

Open or on-request material can shorten the path to inspection, but benchmark readiness still requires the complete configuration, measurement, processing, baseline, and provenance contract. Availability is therefore one input to reconstruction rather than its substitute. These classifications remain a reporting and discoverability snapshot: the review did not contact authors, universally retest links, verify repository preservation, or independently execute the artifacts.

Data and code also support different forms of reconstruction. Open data occur across simulation, laboratory, prototype, computer-vision, and field-data settings [71], [84], [85], [89], [90], while one computer-vision record reported only partial code or model components [71]. Data can support reanalysis without reconstructing the transmitter; code can reproduce

processing without recreating the channel, geometry, or calibration. Hardware-oriented reproduction further needs power references, component settings, clocks, raw traces, and pre-processing choices.

The TQAF reproducibility scores reflect that conjunction: 4 studies were low, 199 adequate, and 3 strong. A strong score requires substantial parameter reporting together with open data or code/model support. It is therefore consistent for 13 studies to report open data while only 3 meet the stronger combined condition.

E. Benchmark Readiness and the Validation Contract

No study reached TQAF benchmark-readiness score 3; 48 were low and 158 adequate. Strong readiness requires an external or common baseline, an open artifact, a directly admissible comparison, adequate validation and reproducibility, and no quarantined claim. This conjunctive rule does not imply that the corpus lacks credible experiments. It shows that none satisfied the full portable-benchmark contract used here.

That contract should be designed into future experiments. The communication and sensing outcomes must come from a traceable configuration whose waveform, hardware, geometry, channel, calibration, and processing settings are preserved. Each performance value needs a stated measurement plane and a clear distinction between raw and derived results. Modality-aware benchmark packages can then combine versioned data, executable processing, machine-readable configurations, and common baselines without forcing unlike physical tasks onto one leaderboard. Field protocols should state which functions operated concurrently, which were replayed or inferred offline, and what changed from the laboratory configuration.

These distributions remain descriptive. They do not show that open artifacts cause better validation, that prototypes cause higher performance, or that field studies are universally more reliable than controlled experiments. Some scores follow mechanically from the TQAF rules, and the corpus was not designed for causal inference. The evidence supports a narrower conclusion: experimental and prototype validation is substantial, but paired-function field evidence, portable benchmarks, and openly reconstructable results remain uncommon under the stated criteria.

VII. ENABLING TECHNOLOGIES, APPLICATIONS, AND 6G NETWORK IMPLICATIONS

Technology and application labels describe different layers of an O-ISAC system. Technology labels identify how a joint signal is generated and transported as well as how it is observed or controlled. Application labels instead express the service conditions that make those mechanisms useful. The 6G frame adds a further burden: a physical mechanism must be connected to a network role and validated service before it supports a readiness claim. Because the technology and application maps are multi-label, their counts indicate within-corpus coverage rather than disjoint families or technical rank. The synthesis therefore follows the chain from physical observables through spatial and infrastructure control to operating requirements and network evidence.

A. From Shared Samples to Physical Observables

The most frequent technology labels occupy complementary positions in that chain. Photonic terahertz or microwave-photonic generation appears in 68 studies, and FMCW or related chirped processing appears in 66. Coherent optical techniques appear in 64 studies, while OFDM or related multicarrier methods appear in 56. These are not four competing waveform choices because photonic generation creates or translates a high-frequency carrier. Chirps map delay or motion into beat frequency, whereas coherent reception preserves phase and OFDM exposes independently controllable time-frequency resources; a single architecture may therefore use all four.

In the reviewed systems, OFDM supports integration through resource placement, with payload tones and pilots providing communication and sensing observables in different proportions. Other reuse points include cyclic prefixes or reserved subcarriers, although the label alone does not show whether sensing relies on payload or known pilots rather than a separate processing branch. Optical modulation constraints and the plane at which SNR is defined also remain decisive. Embedded-pilot fiber sensing and indoor optical positioning illustrate how the same multicarrier mechanism can serve different observables and geometries [65], [91].

Chirped and coherent systems can exploit a different design asset: phase-continuous timing. A returned chirp can reveal range or motion while the transmitted signal carries data, and a coherent front end can preserve small phase changes for distributed sensing as well as demodulation. Photonics-assisted generation extends that timing reference to wideband millimeter-wave or terahertz carriers, as illustrated by full-photonic de-chirping and coherent fiber vibration sensing [57], [82]. That benefit depends on stable lasers and reference distribution as well as sufficient conversion efficiency, receiver bandwidth, and calibration.

In the reviewed photonics-assisted systems, a radiated carrier's communication and sensing paths obey wireless propagation rather than FSO physics. This boundary matters because higher carrier frequency does not guarantee better range, mobility, or weather tolerance. More generally, interfaces between stacked technologies may need to preserve clock and phase while closing the same power and dynamic-range budgets. The transfer point between otherwise mature components can therefore limit the joint system.

B. Spatial Control, Coverage, and Multiplicity

Spatial technologies are less common but alter both link gain and sensing geometry. Beamforming is coded in 13 studies, and optical phased arrays (OPAs) in 11. MIMO appears in 7 studies, while reconfigurable intelligent or optical intelligent surfaces (RIS/ORIS) appear in 2. Beamforming denotes a function, whereas an OPA is one optical steering mechanism; MIMO exploits multiple spatial dimensions, while RIS/ORIS modifies an intermediate boundary. Their counts overlap and should not be combined.

A narrow beam can improve received power and angular selectivity while making acquisition, tracking, and blockage

more consequential. Joint OPA–OFDM design and frequency-comb beamforming show how steering can couple coverage to ranging or imaging [85], [92]. MIMO can instead increase communication streams, sensing diversity, or isolation between targets and paths. Structured optical fields and hybrid MIMO-FSO/FBG systems place that multiplicity in different parts of the architecture [50], [51]. Comparisons must therefore identify which spatial dimensions are independent and how they are calibrated.

The two RIS/ORIS records define a small branch of the evidence map rather than a negative result. One models a mobile FSO path with an optical intelligent surface; the other demonstrates prism-based steering in a photonics-generated terahertz system [93], [94]. Although both redirect energy, their surfaces and propagation laws differ along with their control variables and validation settings, so the evidence cannot yet support a pooled spatial-control claim.

C. Integration, Distributed Infrastructure, and Intelligence

Three mechanism groups move the integration point away from waveform design. Photonic integration appears in 20 studies, fiber distributed acoustic sensing (DAS) in 22, and machine learning or artificial intelligence in 20. Digital twins appear in 2 studies, while 19 studies retain source-specific mechanisms in an “other” category. The latter preserves the long tail without implying that it forms one coherent technology family.

Photonic integration can consolidate generation with reference distribution while combining beam control and detection, thereby reducing duplicated paths. This compactness can concentrate thermal drift and crosstalk while adding packaging and calibration demands. Fiber DAS takes another route: it reuses the transmission medium itself as a spatial sensor. The observable may arise from phase or backscatter and, in some systems, from polarization. Coherent multicore transmission with distributed sensing and operational submarine-cable monitoring illustrate two scales of this infrastructure reuse [36], [95]. In both cases, the governing question is which optical observable can be retained without disrupting carried traffic.

Learning converts those observations into estimates or decisions for event classification and positioning, while other uses include field reconstruction and channel prediction as well as resource allocation. These tasks do not share a meaningful accuracy scale. Fiber diagnosis and federated visible-light positioning cover distributed inference, while wearable optical sensing adds a human-facing task. Together, they expose different labels and ground-truth methods as well as different failure costs [89], [96], [97]. Claims of improvement must therefore retain the training distribution and evaluation shift together with uncertainty and inference cost against a task-specific baseline.

The two digital-twin studies reinforce that need for an explicit decision contract. One models vehicular edge resources, while the other supplies a model-aided physical reference for coherent fiber sensing [98], [99]. “Digital twin” is consequently not a fixed component. Its evidence depends on the represented state and update method, with the mismatch model and consuming decision providing the remaining context.

D. Applications as Bundles of Operating Requirements

Application labels become technically useful only when translated into the target, service role, geometry, timing, and failure constraints they impose. Table VII performs that translation while retaining overlapping corpus coverage.

The common pattern is not one preferred architecture but a shift in the active constraint: mobility emphasizes acquisition and continuity, deployed infrastructure emphasizes nondisruptive persistence, and human or extreme environments add safety, privacy, pointing, packaging, or maintenance. The following 6G subsection therefore treats relevance as framing rather than readiness.

E. From 6G Framing to Network Evidence

The finalized study crosswalk classifies 138 studies (67.0%) as directly relevant to 6G, 64 (31.1%) as inferentially relevant, 1 (0.5%) as weakly relevant, and 3 (1.5%) as not applicable. The direct category is the canonical normalization of source specific codes judged to establish a direct connection to 6G or access functions; it is not equivalent to explicit 6G wording in every paper. Inferential relevance preserves a plausible enabling connection without strengthening the source’s language. The remaining categories preserve cases in which that link is marginal or unsupported. The association is technically plausible because IMT-2030 includes ISAC as a usage scenario [4], while optical technologies can support access, transport, localization, environmental awareness, and high-frequency carrier generation. It remains a framing signal rather than evidence of standard conformance, interoperability, or network-scale readiness.

Explicitly framed beamforming and transport studies [85], [100], technically aligned fiber systems [57], [65], and the weak or not applicable tail [105], [106] illustrate how the rule was applied. These examples distinguish source framing; they do not form a hierarchy of scientific value.

Direct relevance is not a readiness score and cannot replace evidence of maturity, interoperability, implementation cost, or deployability. Inferential relevance likewise does not predict adoption. The distribution supports a bounded connection to future access and transport as well as localization, automation, and environmental awareness; it does not establish compliance with a released standard or conformance to a network interface.

Moving from that frame to network evidence requires the optical or photonic subsystem to occupy an explicit end-to-end role. Access, fronthaul, transport, edge processing, and service control introduce timing, mobility, orchestration, energy, security, and recovery requirements that a standalone waveform cannot resolve. A 6G-facing experiment becomes persuasive when both communication and sensing remain useful under those network variations and the cost of maintaining both is reported.

The cross-layer implication is therefore more useful than the label itself. Waveforms create observables that coherent or photonic hardware preserves or generates; spatial systems place those observables, distributed infrastructure extends their reach, and learning converts them into decisions. Applications

determine which constraints matter. A credible network claim must connect this physical chain to a validated service while leaving standard compliance and deployment readiness as outcomes to be demonstrated.

VIII. DISCUSSION, RESEARCH ROADMAP, AND LIMITATIONS

A. Cross-Cutting Interpretation

The evidence recasts O-ISAC as a shared engineering problem rather than a uniform architecture. Fiber, optical wireless, and photonics-assisted radio systems expose different observables and place the communication–sensing connection at different points. This reading connects architecture-centered syntheses [8], [15] with medium-specific reviews of visible-light and fiber systems [11], [12]. What transfers is the design question: which infrastructure, signal, or information can be shared without making either function unusable? The common question supports a field-level synthesis, while the different physical answers prevent a universal platform ranking.

Within this framework, integration should consequently be interpreted by location and purpose, not by how many sharing labels a design accumulates. A common source can feed a shared waveform whose resources are jointly allocated, so hardware, waveform, and resource categories can describe successive locations in one system. Sharing additional elements may reduce duplicated components and improve coordination, but it can also join power budgets, distortion, synchronization, dynamic range, and failure modes. The relevant comparison is what the shared element enables and which new constraint it introduces.

Metric and tradeoff evidence leads to the same conclusion from another direction. A reported value remains meaningful within its stated task, but it does not travel across studies merely because its unit or label matches. Measurement plane, operating condition, baseline, estimator, and target define what can be compared. Tradeoff points require an equally stable constraint set: rate–resolution, power–accuracy, and coverage–localization couplings do not collapse into one frontier when the underlying modalities and tasks change. Model-centered reviews can still provide bounds and design relations within their stated assumptions [9], [18]; the corpus-level evidence does not convert those relations into one empirical scale. Mechanism-level patterns and bounded case comparisons are therefore more defensible than a leaderboard assembled from headline values.

Validation and artifact evidence expose the practical consequence of that limited portability. The reviewed corpus has a substantial experimental and prototype base, yet end-to-end paired-function evidence remains harder to transfer between laboratories or into deployment. Mature components do not guarantee a mature joint system when interfaces, calibration, or operating conditions are tested separately. Reported artifacts could support cumulative evaluation, but their presence alone does not reconstruct hardware, geometry, and processing. Deployment-centered syntheses broaden the operational record in terrestrial and submarine fiber networks [20], [22]; they do not remove the need for configuration continuity when results

are compared or reproduced. Within this synthesis, the central maturity gap is continuity of evidence across the complete communication–sensing chain.

B. A Research Roadmap Built Around Evidence Contracts

For standards bodies and 6G programme coordinators, the immediate implication is to define reporting profiles tailored to each modality and minimum measurement metadata before proposing performance targets across platforms. Funding calls can reinforce the same goal by requiring reusable condition records, benchmark definitions, and artifact preservation alongside demonstrations. These are enabling policies for cumulative evidence, not recommendations to standardize one waveform, optical medium, or sensing task prematurely.

The evidence gaps become useful only when translated into an experimental decision. Table VIII orders five priorities by dependency and pairs each with a stress test, success criterion, and required artifact.

The dependency chain begins with interpretable observations, proceeds through comparable benchmarks and realistic joint operation, and then makes those tests reusable and extensible to adaptive or network-scale systems. The limitations below delimit the evidence from which this roadmap was inferred.

C. Limitations of This Review

The evidence base reflects a six-source, English-language search covering 1 January 2020 through 22 June 2026; the protocol and deviation history are reported in Section III. Of the sought reports, 58 could not be retrieved, and exact query-to-export mapping was incomplete for two low-yield Taylor & Francis exports. The OSF record was created retrospectively after searching and screening and describes a predecessor corpus. These features can limit coverage and prevent the registration from serving as a prospective constraint on the final review.

The workflow was investigator-supervised and AI-assisted and did not use routine independent duplicate human screening, extraction, appraisal, or adjudication across the full corpus. The safeguards reported in Section III improve traceability but do not replace reviewer independence. TQAF is review-specific and nonvalidated rather than a conventional risk-of-bias instrument. We did not apply GRADE, conduct a formal missing-results or publication-bias assessment, perform a sensitivity analysis, run a meta-analysis, or estimate causal moderators. These choices restrict the strength and type of inference available from the synthesis.

Taxonomy and multi-label coding remain judgment-dependent, particularly for hybrid architectures, and claim counts are not independent effects. Ninety-two metric rows carried forward from the earlier extraction state were conservatively classified as lacking the conditions needed for comparison. Artifact status reflects source reporting at extraction time; external links were not universally retested, and artifacts were not executed under a common protocol. Finally, corpus distributions describe the retained evidence rather than worldwide research activity, market share, or deployment prevalence.

These boundaries do not erase the synthesis; they specify where it can be trusted. The evidence supports conclusions about recurring mechanisms, measurement constraints, and validation gaps within the reviewed corpus. It does not support causal claims, universal rankings, or prevalence estimates beyond that boundary. Keeping those limits visible is what allows the roadmap to identify the experiment still needed rather than merely the topic that should receive more attention.

IX. CONCLUSION

The synthesis identifies integration location and evidence comparability as separate questions. Reuse may occur in hardware, a carrier, a waveform, a resource, a link, joint design, or an application context; none of these labels alone specifies the observable or the constraint transferred between functions. The comparison framework therefore pairs the integration mechanism with the modality, signal path, task, and measurement plane before relating outcomes across studies.

Across the reviewed evidence, communication and sensing were considered within the same studies, and those studies spanned simulation, laboratory, prototype, and field settings; nevertheless, reported performance remains difficult to transfer between studies. Matching a metric name or unit cannot resolve differences in task, measurement plane, operating condition, baseline, and validation setting. Tradeoffs are similarly informative within stable design contexts but do not form a universal frontier across platforms. Laboratory and prototype studies provide a substantial technical base, while paired-function field evidence and portable, reconstructable benchmarks remain uncommon under the stated review criteria.

The next step is therefore to make results transferable rather than simply to accumulate isolated values. Explicit measurement contracts should preserve the conditions that join communication and sensing. Benchmarks need a common reporting core together with modality-specific scenarios, and field protocols must exercise both functions under the disturbances expected in use. Versioned data, code, calibration, and configuration records can then support cumulative evaluation across individual demonstrations. On that foundation, O-ISAC can advance through defensible comparison and cumulative, repeatable engineering evidence.

TABLE V

METRIC EVIDENCE AND COMPARISON CONTRACT FOR 4,779 PRIMARY RECORDS. STUDY COVERAGE IS MULTILABEL; THE FOUR LEADING DOMAINS CONTAIN 4,490 RECORDS AND SMALLER NORMALIZED DOMAINS CONTAIN 289. CONDITIONAL ADMISSIBILITY DOES NOT CREATE A COMMON NUMERICAL POOL.

Panel A—Primary metric domains						
Domain	Records, <i>n</i> /4,779	Studies, <i>n</i> /206	Source metric-family examples	Dominant plane	Principal semantic guardrail	
Sensing	1,816 (38.0%)	203 (98.5%)	Resolution/error; localization or estimation; sensing SNR; detection/classification	Sensing output and ground truth	Resolution, estimator error, bounds, and decision scores differ.	
Communication	1,328 (27.8%)	194 (94.2%)	Data/symbol rate; BER; EVM; OSNR/SNR	Optical link, electrical/digital receiver, or service	Retain payload, coding stage, noise reference, and receiver state.	
Joint	870 (18.2%)	158 (76.7%)	Linked communication–sensing tuples; bandwidth, carrier, distance, resources	Design, joint, or system	A joint point requires documented condition linkage.	
Implementation	476 (10.0%)	64 (31.1%)	Sampling/converter rates; receiver bandwidth; latency; settings	Hardware, configuration, or system	Settings are not intrinsic task performance.	
Other normalized domains	289 (6.0%)	33 (16.0%)	System, security, learning, energy, scenario, waveform, channel, other	Source specific	Normalization preserves source role; it does not authorize pooling.	
Panel B—Semantic comparison contract						
Quantity class	Labels that must not be collapsed	Required definition and accounting	Plane and reference	Required conditions, target, or baseline	Permitted synthesis when aligned	Prohibited shortcut
Rate and information	Symbol, gross/net, aggregate, throughput, capacity, efficiency	Payload, overhead, direction, aggregation, bandwidth	Signaling, decoded link, or service	Frame, modulation, coding, load, resource split	Same accounting and conditions	Rank unlike rates by magnitude
Reliability and error	Pre/post-FEC BER, loss, outage, EVM, Q-factor	Event, coding state, threshold, aggregation	Optical, electrical, digital, packet, or service	Link, receiver/DSP, modulation, coding, baseline	Same event and stage under matched conditions	Replace an error event with a quality proxy
Signal quality and power	OSNR, optical Q, electrical/digital/sensing SNR, launch/received power	Noise bandwidth, estimator, reference, aggregation	Explicit optical, electrical, digital, or sensing point	Detector, filter, bandwidth, gain, power	Within-plane relation or supplied conversion	Move between planes because units are dB
Distance and latency	Span, link distance, target range, coverage; propagation/processing/loop latency	Endpoints, path, timing origin, stages	Geometry, propagation, processing, or system	Topology, target, motion, environment, timing	Same geometry or latency scope	Rank unqualified reach or delay
Sensing resolution and error	Range/spatial resolution, MAE, RMSE, localization error, precision, CRB	Measurand, statistic, estimator, bound, ground truth	Sensing output and stated reference	Target, sampling, calibration, trials, validation	Same task, statistic, and reference protocol	Equate resolution, estimator error, and a bound
Detection and classification	Accuracy, precision, recall, false alarm, detection probability	Classes, averaging, threshold, uncertainty	Decision output and dataset	Split, balance, background, geometry, baseline	Same labels and evaluation protocol	Rank unmatched datasets or thresholds
Joint operating tuple	Communication and sensing values reported together	Condition linkage, resource rule, objective, constraints	Joint/system point	Shared waveform/resource, scenario, baseline, validation	Matched tuple or conditional relation	Pair unlike conditions or invent one ISAC score
Implementation outcome	Bandwidth, sampling, latency, memory, power, component count	Scope, hardware, configuration, version	Device, processing, or system	Load, calibration, algorithm, platform baseline	Matched implementation contract	Treat settings or block count as performance

Notes: The leading domains reconcile as $1,816 + 1,328 + 870 + 476 = 4,490$ and $4,490 + 289 = 4,779$. Study counts are multilabel and must not be summed. The 4,779 primary rows derive from 4,861 governed metric rows after removing 31 context-only and 51 quarantined rows. Units remain source native; curves were not digitized and new values were not derived. FEC: forward error correction; EVM: error vector magnitude; OSNR: optical signal-to-noise ratio; SNR: signal-to-noise ratio; CRB: Cramér–Rao bound.

TABLE VI

REPORTED ARTIFACT STATUS FOR 206 STUDIES AND THE BENCHMARK RECONSTRUCTION CONTRACT. STATUSES REFLECT EXTRACTION TIME; LINKS WERE NOT UNIVERSALLY RETESTED AND ARTIFACTS WERE NOT EXECUTED UNDER ONE PROTOCOL. PANEL B IS NOT A SECOND PREVALENCE DISTRIBUTION.

Panel A: source-reported artifact status

Artifact	Reported status	Studies	Status establishes	Status does not establish	Reconstructability follow-up
Data	Unavailable/not reported	145/206 (70.4%)	No accessible route identified at extraction.	Data absence, refusal, or impossible release.	Recover data, schema, conditions, and provenance.
Data	On request	41/206 (19.9%)	Source states a request route.	Receipt, completeness, license, preservation.	Obtain a versioned snapshot, schema, conditions, and terms.
Data	Open	13/206 (6.3%)	Open status identified.	Coverage, persistence, or rerun fitness.	Verify release, license, schema, checksum, and conditions.
Data	Not applicable	7/206 (3.4%)	Release coded not applicable.	Configuration or result reconstructability.	Preserve rationale and configuration or analytical inputs.
Code or model	Unavailable/not reported	197/206 (95.6%)	No accessible route identified at extraction.	Software absence or impossibility of release.	Supply steps, parameters, dependencies, and an executable substitute.
Code or model	On request	7/206 (3.4%)	Source states a request route.	Receipt, execution, version, or permission.	Obtain exact version, environment, license, and instructions.
Code or model	Partial components	1/206 (0.5%)	Some components identified.	End-to-end pipeline or principal comparison.	Declare missing stages; provide interfaces, parameters, and tests.
Code or model	Not applicable	1/206 (0.5%)	Release coded not applicable.	Hardware, analysis, or measurement reconstruction.	Preserve equations, procedure, configuration, and validation.

*Table VI (continued): Panel B is a controlled contract without a prevalence denominator.**Panel B: benchmark reconstruction contract*

Required element	Source detail needed	Why needed	Establishing evidence or test	Failure if absent
System configuration	Block diagram, interfaces, both tasks, concurrency, validation	Defines connections and comparison	Machine-readable configuration; traceable run to both outputs	Components can be mistaken for one joint point
Hardware and photonics	Parts, transfer functions, bias, power planes, clocks, receiver limits	Recreates budgets and nonidealities	Bill of materials; calibrated settings and reference-plane measurements	Power, distortion, and dynamic-range differences remain unexplained
Geometry, channel, environment	Topology, distance, target, channel, mobility, atmosphere, illumination, traffic	Recreates scene and operating condition	Scenario plus measured or validated channel/environment record	Performance cannot be separated from the setting
Calibration	Timing, phase, alignment, references, ground truth, interval, uncertainty	Maps observations to quantities and functions	Protocol, reference data, and uncertainty record	Bias, drift, and uncertainty remain unknown
Waveform and resources	Modulation, sampling, timing, bandwidth, carrier, power, pilots, coding, allocation	Recreates excitation and shared resource	Generation file or parameter manifest reproducing the signal	Operating point and resource split remain ambiguous
Processing and algorithms	Preprocessing, estimator/model, hyperparameters, seeds, split, dependencies, order	Reproduces samples-to-output transformation	Executable declared environment with reference input/output	Result is inspectable but not repeatable
Metrics and reference planes	Formula, unit, denominator, plane, gross/net, error stage, baseline, aggregation	Preserves each value's meaning	Worked calculation and baseline check from outputs	Identical labels can denote different quantities
Raw/intermediate data	Traces, labels, ground truth, splits, exclusions, transformations, figure/table links	Enables audit, reanalysis, and disagreement localization	Checksummed schema, lineage, and reproducible split	Final plots alone cannot test processing choices
Version, license, preservation	Release ID, dependencies, license, checksums, archive, change log	Identifies reusable state	Licensed persistent archive and validated manifest	Link drift, version ambiguity, or permission blocks reuse

TABLE VII

APPLICATION DOMAINS AS OPERATING-REQUIREMENT BUNDLES. COUNTS ARE MULTILABEL ACROSS 206 STUDIES AND DESCRIBE REVIEWED EVIDENCE, NOT MARKETS, DEPLOYMENT PREVALENCE, OR RANK; REQUIREMENTS REMAIN QUALITATIVE UNLESS A COMMON SOURCE-GROUNDED DEFINITION EXISTS.

Group and domain	Studies	Sensing role	Service and geometry	Operating requirements	Constraint and interpretation boundary	Evidence
Network/mobility: 6G/access	100/206 (48.5%)	Channel, target, position, or environment for awareness	Access, fronthaul, transport, or control; indoor/outdoor, distributed/mobile	Timing, orchestration, continuity, paired service	Scale and interfaces are source specific; service frame, not the exclusive 6G relevance code or readiness	[50], [100]
Network/mobility: vehicular	41/206 (19.9%)	Range, velocity, pose, temperature, channel state	V2X, in-vehicle payload/control; moving roadside or in-vehicle terminals	Acquisition, tracking, Doppler tolerance, continuity	Safety/latency thresholds vary; label spans paths, tasks, and validation	[67], [101]
Network/mobility: optical access	32/206 (15.5%)	Channel, event, or fiber state	Access, fronthaul, or transport; guided access topology	Live traffic, reach, synchronization, nondisruptive sensing	Service impact and interfaces are source specific; topology differs from broad 6G framing	[65], [89], [100]
Network/mobility: indoor positioning	25/206 (12.1%)	User/robot position and orientation	Downlink payload/control; indoor line of sight, reflections, mobile receivers	Calibration, coverage, update continuity, orientation	Privacy/safety not uniformly coded; accuracy requires matched geometry	[44], [91]
Network/mobility: datacenter	4/206 (1.9%)	Channel or link state	Short-reach high-rate data; dense guided fiber	Continuity, low disruption, calibrated references	Facility constraints vary; four studies do not establish readiness	[65]
Infrastructure: environmental monitoring	55/206 (26.7%)	Vibration, weather, ocean, temperature, hazards	Operational transport/backhaul; distributed fiber, exposed free space, or subsea	Persistence, localization, traceable ground truth	Maintenance/service disruption are source specific; observables and media differ	[36], [89], [102]
Infrastructure: smart infrastructure	41/206 (19.9%)	Structural, motion, or event state	Monitoring transport, alerts, control; distributed built infrastructure	Continuous observation, calibration, resilience	Access/failure costs vary; membership is not deployment maturity	[50], [89], [103]
Infrastructure: industrial	34/206 (16.5%)	Process, strain, temperature, motion, or position	Control/data/last mile; factory, fiber, free space, wearable, or robotic	Continuity, bounded delay, calibration	Safety thresholds are source specific; tasks lack one scale	[50], [97]
Infrastructure: security/surveillance	26/206 (12.6%)	Intrusion, anomaly, event class, or location	Alert/monitoring transport; perimeter, indoor, or distributed	Coverage, false alarms, timely alerting	Privacy/failure costs vary; label is not attack robustness	[89]
Human-facing: healthcare	8/206 (3.9%)	Vital signs, activity, or gesture	Wearable data/control; on-body or near-body ambient-light setting	Motion robustness, physiological ground truth, safe illumination	Privacy, contact, wearability, and safety; membership is not clinical validity	[35], [97]
Extreme: aerospace	20/206 (9.7%)	Wavefront, atmosphere, range, or motion	Satellite-ground data/control; long moving free-space path	Pointing, Doppler, atmosphere, delay	Availability/maintenance vary; simulation or laboratory framing is not flight readiness	[104]
Extreme: underwater	2/206 (1.0%)	Temperature, motion, coastal/seabed events	Subsea optical/fiber transport; water column or cable	Absorption/scattering or long-route calibration; packaging	Difficult access; wireless and guided paths have different channel laws	[102], [103]
Residual: other	15/206 (7.3%)	Source-specific range, imaging, object, or channel task	Source-specific payload/control; unmatched or emerging context	No category-wide threshold	Constraints not uniformly coded; normalization fallback, not absence of an application	[94]

Note: Counts are multilabel and must not be summed.

TABLE VIII

EVIDENCE-LINKED RESEARCH ROADMAP. EACH PRIORITY TRACES TO A REVIEW RESULT AND HAS A TESTABLE SUCCESS CRITERION. PROPOSED ACTIONS WERE NOT EVALUATED BY THE REVIEW AND MUST PRESERVE MODALITY-SPECIFIC PHYSICS.

Priority	Gap and evidence trace	Required action	Stress test, baselines, and controls	Success criterion	Artifact	Risk
Measurement contract	Labels and units often lack portable context. Section V: 4,779 metric records, 118 conditionally admissible, none unconditional	Retain task, plane, unit, condition, baseline, validation, and locator	Fiber: topology/traffic; FSO: geometry/pointing/atmosphere; VLC/LiFi: illumination/field of view/ambient/orientation; photonic THz: optical front end/wireless path. Check native definition, raw/derived value, and a common reference where meaningful	Independent reader reconstructs quantity, denominator, and condition	Machine-readable result, schema, provenance	First dependency; omissions propagate false equivalence
Modality-aware benchmarks	Common names do not create a physical scale. Sections II and V: comparison is conditional on matched context	Share a reporting core while preserving modality physics and task ground truth	Repeat one declared task under common calibration; compare reference design and independent implementation while varying environment and calibration	Independent groups repeat the relative finding within the contract	Scenario, calibration, baseline, ground truth, evaluation protocol	Requires row 1; overgeneralization erases native meaning
Joint operation under disturbance	Most relations are conditional. Section V: 371/402 substantive relations; Section VI: 6/12 highest-tier studies met the paired-function field rule	Vary the shared resource and report both functions concurrently under load	Fiber: traffic and topology; FSO: mobility/blockage/weather/pointing; VLC/LiFi: motion/orientation/ambient; photonic THz: front-end drift/wireless variation. Add single-function baselines and vary power, time, bandwidth, wavelength, subcarrier, polarization, beam, or processing	Repeatable joint region retains an identifiable limiting mechanism across conditions	Paired traces, condition log, calibration, single-function baselines	Requires rows 1–2; replay or unmatched conditions break continuity
Experiment-defining artifacts	Access rarely supplies the reconstruction chain. Table VI: data open in 13/206; code/model partial in 1/206, otherwise unreported, on request, or not applicable	Preserve signal, hardware, geometry, calibration, data, software, environment, license, and version	Rerun the principal comparison and one modality-relevant condition against the declared baseline; document synthetic substitutes	Independent rerun reproduces the comparison and evaluates the added condition	Persistent versioned package, manifest, checksums, license, rerun report	Requires rows 1–3; an open link alone does not ensure reuse
Intelligence, scale, and security	Adaptive/network labels do not establish robustness or readiness. Section VII: counts overlap and 6G relevance is framing, not conformance	Expose shift, attack, overhead, latency, compute, energy, multiuser interaction, and fallback	Shift target, channel, traffic, geometry, or calibration; inject suitable attack or component failure. Compare nonadaptive/single-user and single-function references; vary load and attack strength	Joint service persists under a declared shift/attack with an explainable degradation boundary	Shift/threat model, controls, resource accounting, decisions, fallback trace	Earlier contracts required; hidden control cost or unsafe fallback can invalidate service

COMPETING INTERESTS

The authors declare no competing interests.

DATA AND MATERIALS AVAILABILITY

The structured study and report lineage, exclusion, metric, tradeoff, TQAF, evidence body, search, protocol, and data dictionary records supporting this review are provided as electronic supplementary material with this article. Publisher full texts, licensed database exports, credentials, private notes, and internal tool artifacts are not redistributed.

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OpenAI Codex, identified as GPT-5 in the project records, was used throughout article preparation, including the abstract, Sections I–IX, tables, and reporting declarations, to support record preparation grounded in sources, language drafting, structural revision, consistency checking, and quality assurance implemented with code. Google Gemini supplied secondary cross checks during extraction from PDFs; its exact model and version could not be reconstructed from the retained audit trail [40], [41]. AI outputs were treated as proposals or structured evidence records rather than independent reviewer votes. One investigator made the final review decisions, and responsibility for the article remains with its human authors.

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